

FANUC America Corporation R-30*i*B and R-30*i*B Plus Controller MULTI ARM Controller Option Manual

MAROBMA8203131E REV B

Version V8.20 and later

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FANUC America Corporation
3900 W. Hamlin Road
Rochester Hills, MI 48309-3253

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FANUC America Corporation
3900 W. Hamlin Road
Rochester Hills, MI 48309-3253

www.fanucamerica.com

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Patents

One or more of the following U.S. patents might be related to the FANUC America products described in this manual.

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VersaBell, ServoBell and SpeedDock Patents Pending.

Conventions

This manual includes information essential to the safety of personnel, equipment, software, and data. This information is indicated by headings and boxes in the text.

**Warning**

Information appearing under **WARNING** concerns the protection of personnel. It is boxed and in bold type to set it apart from other text.

**Caution**

Information appearing under **CAUTION** concerns the protection of equipment, software, and data. It is boxed to set it apart from other text.

Note Information appearing next to **NOTE** concerns related information or useful hints.

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Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system-including the design, implementation, and debugging of application programs-must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and **not** replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.

- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.
- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points* . Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.
- Before teaching, visually inspect the robot and *work envelope* to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.



Warning

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:



Warning

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.
- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that an applicator bell cup can continue to spin at a very high speed even if the robot is idle. Use protective gloves or disable bearing air and turbine air before servicing these items.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.



Warning

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.

- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

**Warning**

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

Note Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

MULTI ARM

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1.1 MULTI ARM FEATURES AND BENEFITS

MultiARM hardware allows you to program and control up to four robot arms plus positioners (up to 72 axes total) with a single controller (having a single main CPU) and teach pendant. MultiARM provides close coupling of all robots, positioners, index devices, and extended axes motion through the single controller.

See [Figure 1–1](#) through [Figure 1–4](#) .

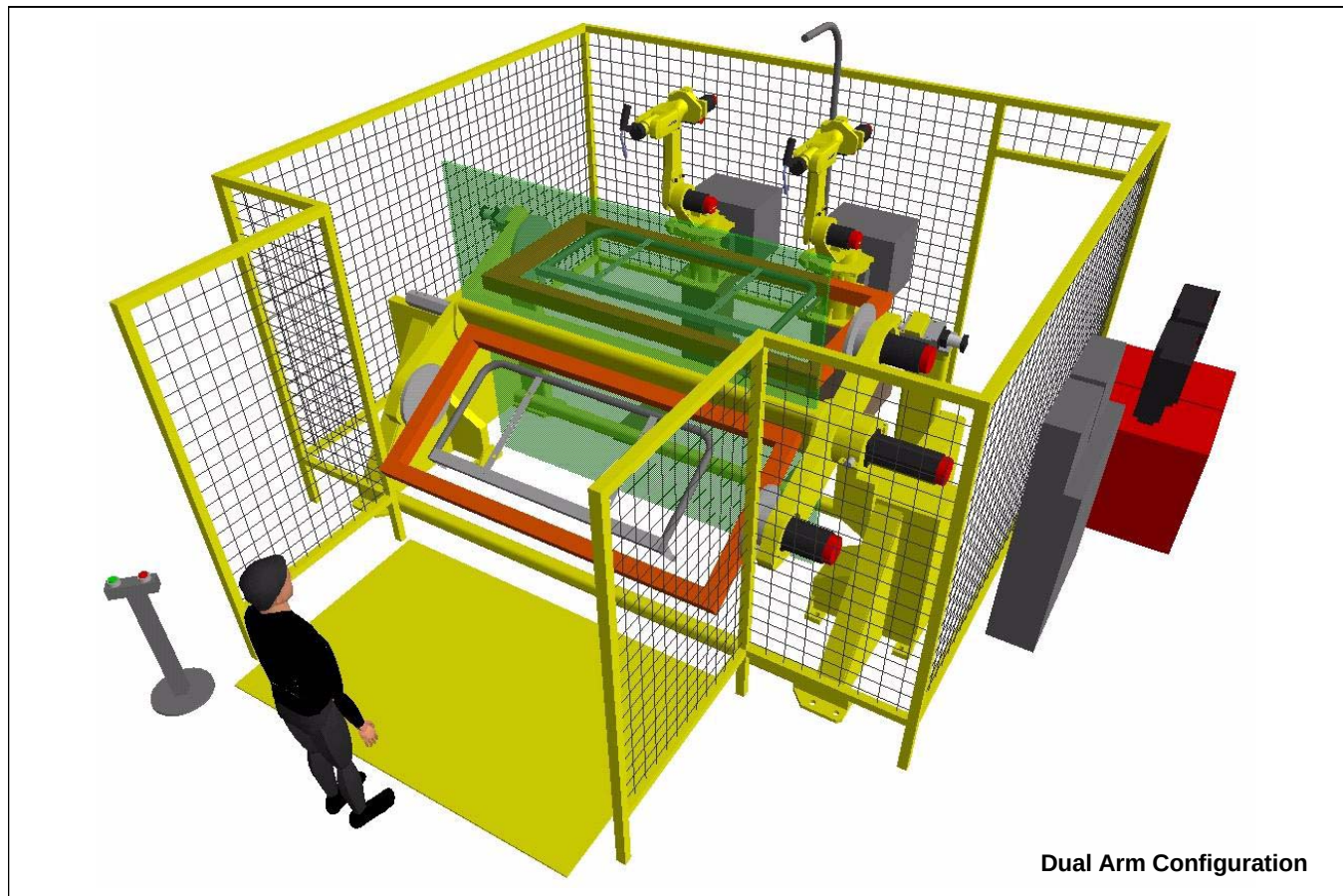
Programming choices allow you to have:

- Zero to eight motion groups in a single program.
- Zero to eight motion groups executing simultaneously in independent programs
- Independent UOP control is available for five programs.
- Up to four robots controlled by the same program.
- Robots and positioners paired as independent work cells.
 - executing multi-group motion programs simultaneously via multi-tasking.
- Up to four robots moving relative to a single positioner in a single program.
- Multiple sets of robot(s) and positioner(s), with independent program execution.

Programming is simplified since no handshaking is required to insure that the robots do not collide while performing their tasks. Cell Interface is reduced: External I/O is not required to coordinate the motion of the multiple robots, therefore work cell integration, programming effort/time, and overall cell cost are reduced. When you run programs, the single controller posts errors that occur on any robot arm. When a single-group or multi-group program has a local alarm (scope is limited to that program only) then all groups controlled by that program will stop; error recovery is required on that program only and other multi-tasking program execution continues. When any executing program has a global alarm, all executing programs are stopped to prevent damage to the robots, equipment, and positioners.

Robot Isolate Function (included in MultiARM Package, R671) - This feature allows motion groups to be locked out so that program testing can be isolated to only the required motion groups in multi-group programs. For example, all robots except one could be isolated to work on welding parameters for the active robot.

Figure 1–1. MultiARM Cell with Ferris Wheel Positioner



The following limitations apply to Multi-Arm.

- Common Style Select does not support concurrent execution of teach pendant programs.

The following options are not supported on Multi-Arm Systems.

- Tracking options (AVC, SERVO ROBOT, LINETRACKING)
- RTCP
- Bump Box option

Note Error Recovery supports five instances of error recovery; this allows error recovery for up to five independently executing programs.

1.2 MULTI ARM SOFTWARE, SYSTEM CONFIGURATIONS, AND PROGRAMMING METHODS

MultiARM hardware allows you to use up to eight motion groups to control up to four robot groups and part positioners. Up to 72 total axes are supported by the robot controller. The hardware and software products combine to form a very flexible and powerful system.

1.2.1 MultiARM Software

MultiARM hardware configurations use the MultiARM system software to provide robot control and multi-equipment control for arc welding applications.

Before you can use MultiARM systems, you will need to know how to use several software features. This section outlines the software features for MultiARM, and also the supported options for these software packages.

1.2.1.1 MultiARM System Software

All MultiARM systems have the following basic software:

Table 1–1. MultiARM Software

FANUC Software	FANUC Part Number
MultiARM Package	R671 (This package bundles most basic software options required to operate a MultiARM system)
Multi UOP Interface (this replaces Multi-Robot Control J605)	J964 (Included in R671)
Robot Isolation	J544 (Included in R671)
Tracking Jog	R624 (Included in R671)
Process Sync	R649 (Included in R671)
Multi-Group Motion	J601 (Included in R671)
Constant Path	R663 (is required)
Triple/Quad Arm	R673 (Required when 3 or more robot arms are installed)

1.2.1.2 Common Options for MultiARM Software

The following options can be incorporated into MultiARM systems. Note that some options do not support all features of the option when installed in MultiARM systems.

Table 1-2. Common Options for MultiARM Software

FANUC Software	FANUC Part Number
ASCII Upload	R507
Bump box	R516
iRCalibration Vision Frame Shift	J991
Coord-Motion Plus	R669 (includes Coordinated Motion and iRCalibration Vision MultiCal J993)
Data Monitor (with limitations) High data sampling rate may increase CPU loading to unacceptable levels.	J675
DeviceNet Interface	J753
Domain Name Serv.	J755
EGD Ethernet I/O (with limitations) High transmission frequency and or large number of I/O will increase CPU loading and impact other Ethernet communication.	R535
Error Code Output	J527
Error Recovery (with limitations)	J664
Ethernet CBR	J717
Explicit Msg. Int.	R541
Extended Axis Control (with limitations)	J518
Extended Error Log	R542
Fronius Eq Lib (with limitations) Manual setting required using Devicenet interface.	R653 (Arc only)
FTP Interface	J716
Filtered Memory Device.	R545
Gravity Compensation	J649
HMI Device (SNPX)	R553
Intelligent Interference Check (with limitations) May need to reduce elements in robot model when CPU reaches unacceptable levels.	R759
Internet Conn/Custo	R558
Collision Guard Package	J684
Lincoln Eq Lib	R661 (Arc only)
Local Stop (with limitations) May require additional I/O hardware	J806
Menu Utility	R577
Miller Eq Lib (with limitations) Manual setting required using Devicenet interface.	R661 (Arc only)
iRCalibration Vision MultiCal	J993 (included by Coord-Motion Plus R669)

Table 1–2. Common Options for MultiARM Software (Cont'd)

FANUC Software	FANUC Part Number
Multi Equipment	J617
PC Interface	R641
Password Protection	J541
Prosync Fctn (with limitations) Can use multiple instances but of same type, either ProcSync or ProcStart/ProcEnd in a program.	R649 (included in MultiARM Package, R671)
Remote TCP (with limitations) - Single group program when basic RTCP is used - Multi-group program, only one pair of robots DRTCP - If COORD option is loaded, RTCP will be DISABLED automatically.	J624
iRCalibration Vision Master	J992
ServoTorch (with limitations) Maximum 2 ServoTorch	J691
Basic Interference Check	R761
iRCalibration Vision TCP	J990
TorchMate	R639
Touch Sensing (with limitations) Refer to Section 2.1 .	J536
Tracking Jog	R624 (included in MultiARM Package, R671)

Note Many of the limitations are noted in this manual. Some limitations are based on CPU loading, memory concerns, or compatibility with other loaded options. Contact FANUC America Corporation regarding software option limitations in MultiARM configurations.

1.2.1.3 Robot Models Supported in MultiARM Systems

Even though most FANUC robots are either supported or capable to be supported in a MultiARM system, some exceptions or restrictions applies, including (but not limited to) the robots listed in [Table 1–3](#) .

Table 1–3. Some Restrictions on Robot Models in a MultiARM system

FANUC Robot Model	Restrictions
M2000iA series robots	only up to 2 robots / controller
M-900iA/200P	only up to 2 robots / controller

Note Contact FANUC America Corporation regarding robot configurations for your system.

1.2.1.4 Extra Group Devices Supported in MultiARM Systems

Table 1–4. Supported Extra Group Devices

FANUC Robot Model	FANUC Part Number
General Positioner	H874
AM Index Device (Does not support Coordinated motion)	H636
Independent Axis (Does not support Coordinated motion)	H895

Note Contact FANUC America Corporation regarding configurations for your system.

1.2.2 MultiARM Software Features and Setup Overview

The following table outlines the basic MultiARM software components.

Table 1–5. MultiARM Software Features

ITEM	DESCRIPTION
Group Motion Control	A motion group is a collection of motors (and associated hardware) that are treated as a single entity for motion. The position data for a motion group contains data for each axis including any auxiliary axis attached to the group. A robot is an example of a motion group. Single and multi-axis positioners are another form of motion group. Refer to the “Turning On and Jogging the Robot” chapter of your application-specific <i>Setup and Operations manual</i> for information about how to select motion groups.
Coordinated Motion	Coordinated motion is a motion control option that defines and controls motions of a follower group in the frame of a leader group. The main purpose is to control TCP motion at the programmed speed and motion linearity with respect to the moving leader frame. MultiARM can be used with the FANUC standard Coordinated Motion in independent “robot + positioner” workcells. Each leader-follower pair is contained in its own program, which can be simultaneously executed via multi-tasking. Refer to the <i>Coordinated Motion manual</i> to set up Coordinated Motion. Up to four robots can also be coordinated to a single positioner in one program if the robot paths are identical or mirrored.
Multi-Equipment	The multi-equipment option supports multiple process hardware attached to a single robot or to multiple robots. This requires control hardware, a setup menu for each equipment, and a means to toggle between equipment. For multi-robot cells, this typically also involves coupling an equipment to a robot motion group. Equipment coupling to a motion group insures that the proper weld equipment is operated and the schedule data for that equipment is applied when a WeldStart instruction executes. Up to four equipment are supported in ArcTool MultiARM systems.

Table 1–5. MultiARM Software Features (Cont'd)

ITEM	DESCRIPTION
Weld Start Synchronization (included feature in Multi-equipment) (ArcTool only)	Weld Start Synchronization is a component of Multi-Equipment that allows multiple robots in the same program to start welding at the same time. Motion of any robot is dependent on all synchronized equipment successfully starting. In cases where a fault has occurred, the arc re-start is performed only after the robots have returned to the fault position using the Return to Path feature. WELD START (for schedule change) and WELD END instructions can be executed simultaneously, or asynchronously.
Multi UOP Interface	Multi UOP Interface J964 (included in MultiARM Package R671) provides independent UOP interfaces to support control of up to five programs independently. The number of program control UOP interfaces is user configurable. • All motion groups can be contained in a single program. Motion is synchronous for all groups. This only requires a single UOP interface for control. • Multiple programs using the MASH "family of programs" structure can contain from 0 or more motion groups that are synchronized from an execution standpoint; "program select" selects all family programs to be selected, "program execution" causes all family programs to execute. Motion is asynchronous across all programs. This only requires a single UOP interface for control. • In MultiARM systems, a single arm can be controlled to move by itself while the other arm remains in a fixed position. Multi UOP Interface provides independent control through additional signals in the UOP interface.
Robot Isolation	This feature provides robot arm lockout through switches on the Standard Operator Panel (SOP). During point touchup or training, the "isolated" groups that are in the edited program will not have their position data altered.

To use MultiARM features you must:

- Set up the robot controller
- Master the robots and positioners or index devices. You could also use iRCalibration Vision Master Option for robot mastering and calibration.
- Set the robots' tool center point (TCP) data. You could use iRCalibration Vision TCP Set Option for setting Robot Tool Center Point.
- Set up MultiARM Coordinated Motion. Traditionally, one uses coordinated motion calibration "Four Point method" for robot-to-robot calibration for HandlingTool installations. For higher accuracy, it is highly recommended to use MultiARM iRCalibration Vision MultiCal Option (which is included in the Coord-Motion Plus Option), especially for multiple robots handling the same part simultaneously.
- Set up Coordinated Motion pairs between robots and positioners. Traditionally, one uses Coordinated motion calibration "three point method." For higher accuracy, it is highly recommended to use MultiARM iRCalibration Vision Multi-Cal Option (included in the

Coord-Motion Plus Option), which could be used to calibrate both robot-to-robot and positioner-to-robot.

- Set up process equipment
- Set up installed options
- Write programs that control your robots and positioners

1.2.3 MultiARM System Configurations

You can configure a MultiARM system in the following ways:

- MultiARM hardware with no additional groups (see [Figure 1-2](#))
- MultiARM hardware with index device groups (see [Figure 1-3](#))
- MultiARM hardware with positioner groups (see [Figure 1-3](#))
- MultiARM hardware with positioner and index device groups, such as Ferris wheels, H-frames, and so forth (see [Figure 1-4](#))

Figure 1-2. Example of MultiARM with Two Workpieces

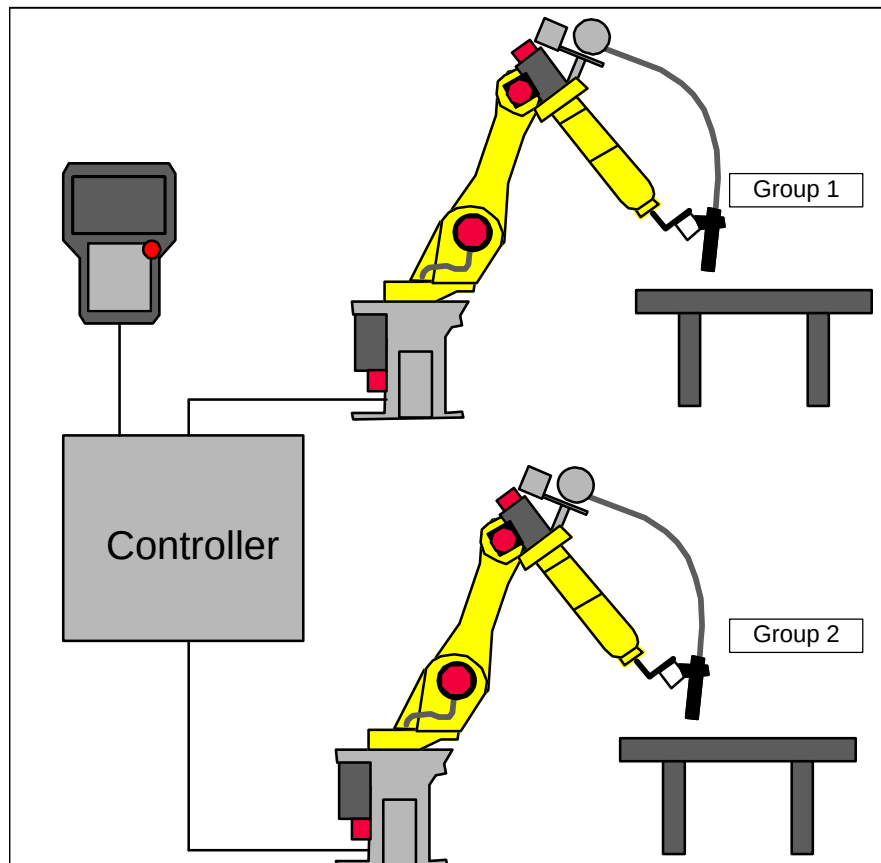


Figure 1-3. Example of MultiARM with Two Single-Axis Positioners

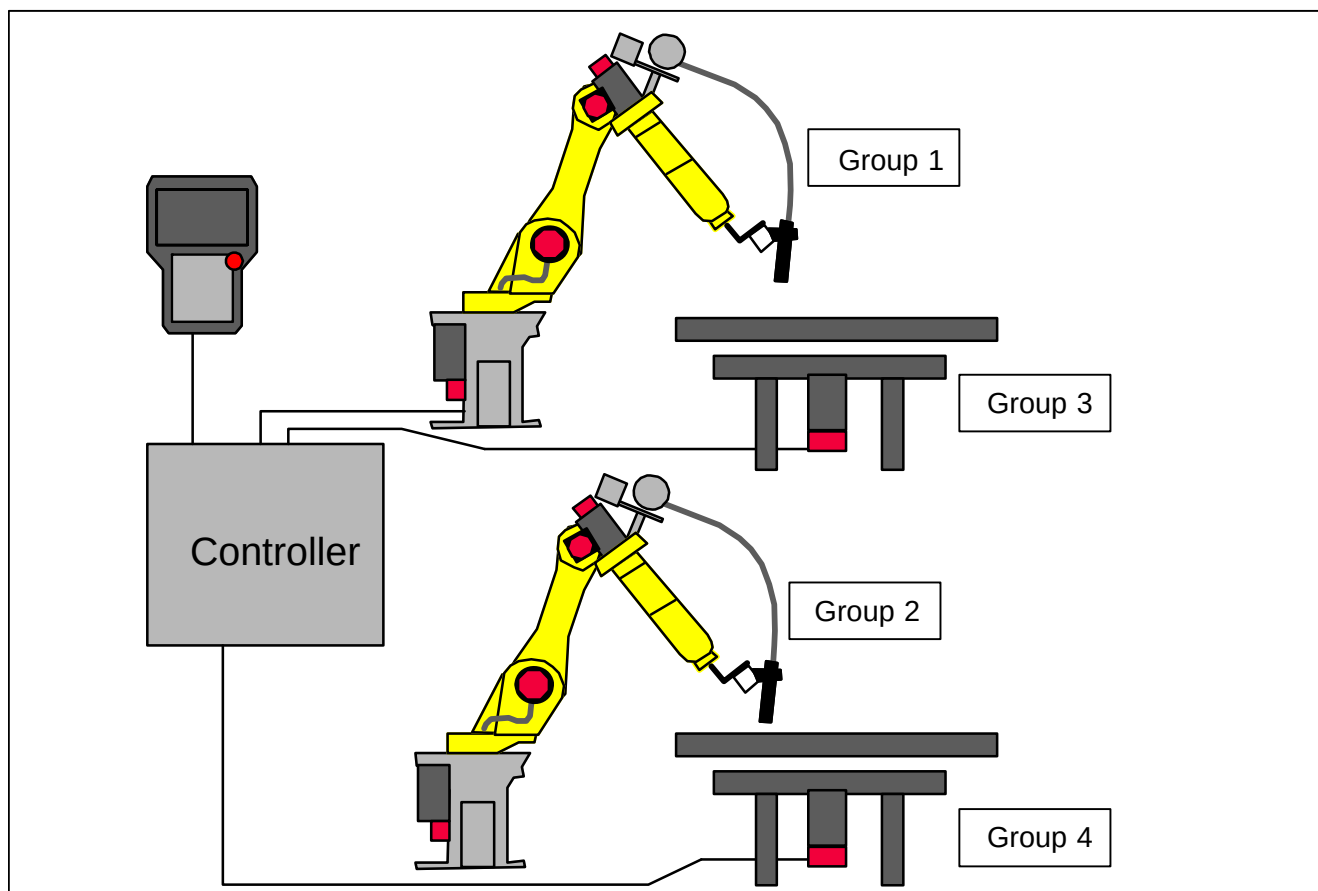
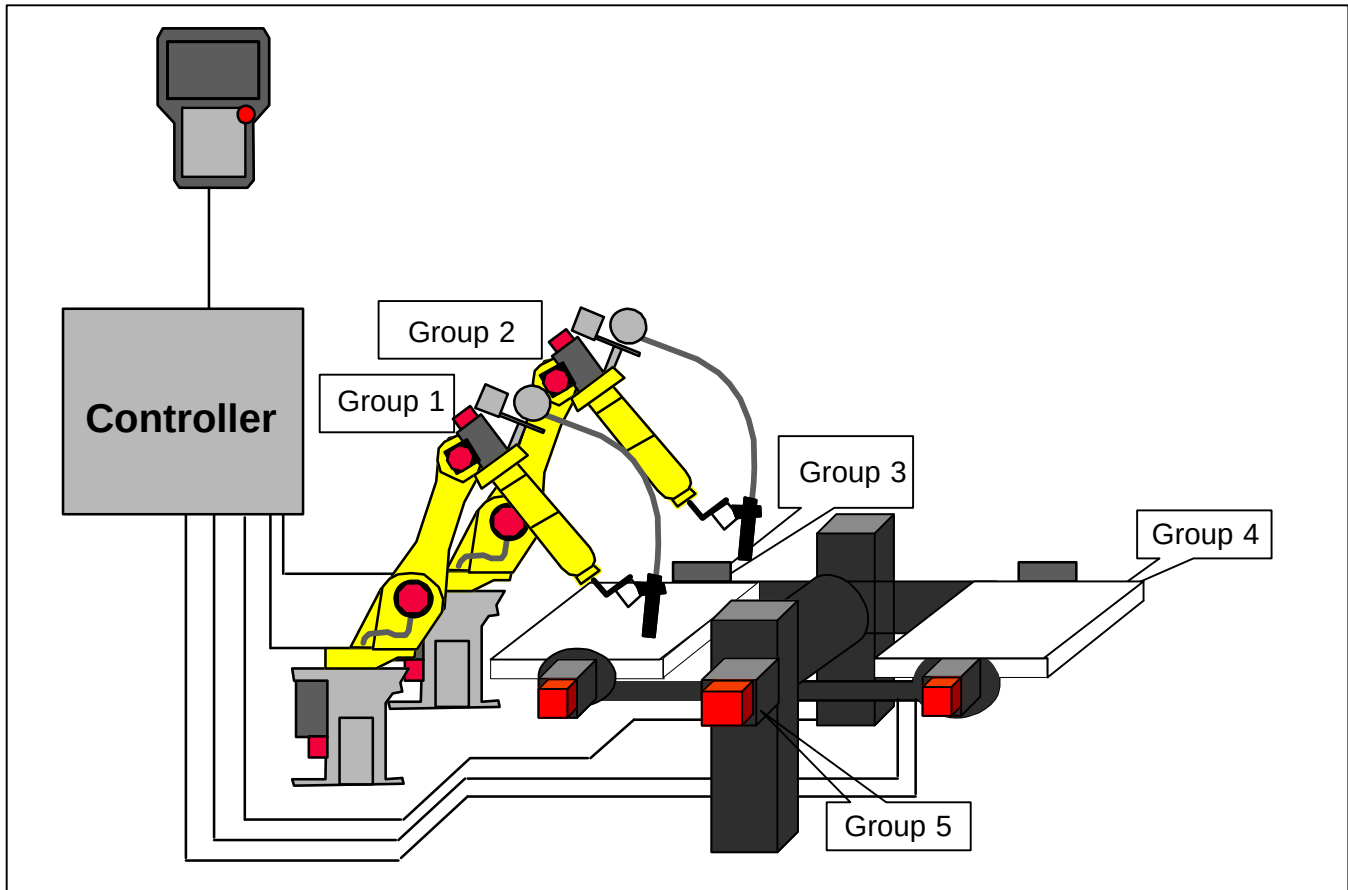


Figure 1–4. Example of MultiARM with Three Additional Groups



1.2.4 MultiARM Tracking Jog

1.2.4.1 Overview

The MultiARM Tracking Jog feature simplifies the teaching of MultiARM programming. Major features of this system include the following:

- Coordinated motion setup is not required for MultiARM Tracking Jog.
- The leader and the follower robot(s) issue the same amount of incremental motion for translation and rotation.
- The system provides two MultiARM jog modes:
 - Track Jog

- Mirror Jog

1.2.4.2 Benefits

Major benefits of MultiARM Tracking Jog include:

- Simplified MultiARM program teaching, including the following:
 - Multiple robot arms are working on one symmetrical part.
 - Multiple robot arms are working on different parts that are identical.
 - Two robot arms are working on different parts that are mirror images of each other.
- During track jog mode, each robot arm jogs with respect to its own active frame. For example, it rotates with respect to its own rotation axis and center.
- During mirror jog mode, the follower robot jogs while corresponding to the leader robot's direction. In this case, the leader robot jogs as a normal right hand robot while the follower jogs as a left hand robot. The mirror plane can be configured by \$track_jog.\$mirror_pln, 0 for Z-X plane, 1 for X-Y plane, 2 for Y-Z plane.

1.2.4.3 Limitations

Limitations of MultiARM Tracking Jog include the following:

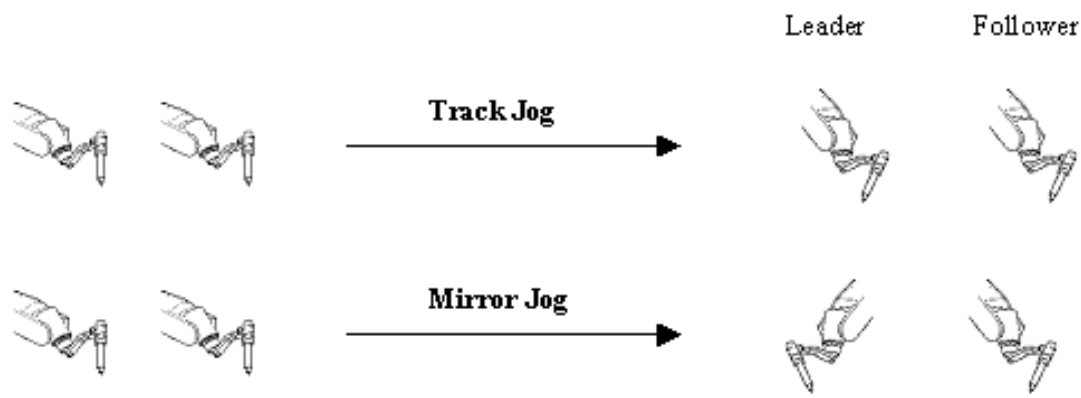
- You are responsible for installing and configuring the required MultiARM hardware and software.
- The MultiARM Tracking Jog feature is provided to jog robot arms only, it will not jog a positioner.
- MultiARM Tracking Jog currently supports the following coordinates:
 - World
 - User
 - Tool
 - Jog
 - Joint

Note MultiARM Tracking Jog does not support Path Jogging, Leader Frame Jog, or Remote TCP Jog.

1.2.5 Using MultiARM Tracking Jog

During track jog mode, the follower robot tracks the motion of the leader robot and each robot arm jogs with respect to its own active frame. For example, it rotates with respect to its own rotation axis and center. See [Figure 1-5](#) .

Figure 1-5. Track Jog and Mirror Jog Examples



During mirror jog mode, the follower robot mirrors the motion of the leader robot and the follower robot jogs while corresponding to the leader robot's direction. In other words, the leader robot jogs as a normal right hand robot while the follower jogs as a left hand robot. For example:

Table 1-6. Leader Robot and Follower Robot during Mirror Jog

Leader	Follower
x	x
y	-y
z	z
w	-w
p	p
r	-r

The mirror plane can be configured by `$track_jog.$mirror_pln`, 0 for Z-X plane, 1 for X-Y plane, 2 for Y-Z plane.

1.2.6 Tracking Jog Modes

There are at least two tracking jog modes: track jog (T12) and mirror jog (M12). Each mode can be activated using the FUNCTION menu. The default setting is Group 1 (G1). Refer to [Table 1–7](#) for a description of these modes.

Table 1–7. Tracking Jog Mode Descriptions

Mode	Description
G1	Group 1
T12	Track Jog Mode – Leader robot 1, Follower robot 2
M12	Mirror Jog Mode – Leader robot 1, Follower robot 2
G2	Group 2
T21	Track Jog Mode – Leader robot 2, Follower robot 1
M21	Mirror Jog Mode – Leader robot 2, Follower robot 1

Toggling Between Modes

1. Press FCTN.
2. To select **Track Jog Mode**, select TOGGLE TRACK JOG.
3. Press ENTER and you will see the following at the top of the screen.

```
SRVO-003 Deadman switch released
CUR LINE 0 T2 ABORTED E1 T12 WORLD
```

4. To select **Mirror Jog Mode**, press FCTN again.
5. Select TOGGLE TRACK JOG.
6. Press ENTER and you will see the following at the top of the screen.

```
SRVO-003 Deadman switch released
CUR LINE 0 T2 ABORTED E1 M12 WORLD
```

If more than 2 robot arms are installed in a controller, more tracking jog modes will be available. For example, for a controller installed with 3 robots in groups 1, 2, and 3, the following tracking jog modes will be available when group 1 is the leader:

Table 1–8. Mode Descriptions: 3 Robots in Groups 1, 2, and 3

Mode	Description
G1	Group 1
T12	Track Jog Mode – Leader robot 1, Follower robot 2
T13	Track Jog Mode – Leader robot 1, Follower robot 3
T123	Track Jog Mode – Leader robot 1, Follower robots 2 and 3
M12	Mirror Jog Mode – Leader robot 1, Follower robot 2
M13	Mirror Jog Mode – Leader robot 1, Follower robot 3

Note Permutations with G2 as leader and G3 as leader are not shown but are available.

1.2.7 MultiARM Programming Methods

The following programming methods can be applied to MultiARM configurations depending on the hardware in the system, the software options, and the manufacturing methods required:

- **Type 1** (Multi Tasks) - Single group programs are used. Each robot program controls a single robot and one equipment. Robots execute independently. Simultaneous execution of two or more robot programs is supported by multi-tasking using multiple UOPs (see Multi-UOP Interface) or single UOP (see MASH)
- **Type 2** (Single Task) - Multi-group programming that includes all robots in the same program.
- **Type 3** (Multi Tasks) - Single robot and positioner are used to manufacture a part. A second program using another robot, and positioner if installed, can execute simultaneously. Standard coordinated motion can be used in each of the robot /positioner programs.
- **Type 4** (Single Task with Coordinated Motion) - Two or more robots are combined with a single positioner in one program. These programs can execute standard coordinated motion with some restrictions. Robot-to-robot coordinated motion is also supported, and additional CD pairs can be defined.

Some of the programming methods require specific program structures using multiple programs. For more information on programming implementation, refer to [Section 1.16](#).

Please review the following programming methods and determine which method best fits the required process. **For Multiple Tasks, it is important to determine the program selection / execution method, and the number of UOPs required, as it will affect the following SETUP for your system:**

- \$MULTI_ROBO.\$NUM_PROG: number of programs / UOPs (up to 5), default is 1.
- \$MULTI_ROBO.\$PROGC[n]: defines which groups are intended for PROGRAM#n (n: UOP or PROGRAM number, n =< 5)
- Enable or Disable MultiARM SHELL (MASH)

Please refer to [Table 1–16](#) for details on these two settings

Programming Method	Typical Number of UOPs required	Description	Advantages
Single Task	One	A program where all motion groups are included in the program group mask	• Simplest way to ensure that all robots and positioners stay synchronized. • Teach one point and all groups are recorded • Ideal for symmetrical paths with multiple robots
Single Task with Coordinated Motion	One	A program where all groups are in the same program, with the addition “COORD” motion option that makes the “Follower” group(s) moves relative to the “Leader” group.	• Allows robots to execute a path “relative” to the Leader group to maintain orientation and path speed • The Leader group can be either a Positioner or a Robot • Ideal for symmetrical paths with multiple robots

Multiple Tasks (using Multi UOP Interface)	Multiple (≤ 5)	• Multiple independent programs, e.g. one for each robot, or positioner. • Each program is selected / executed by a separate UOP (up to 5)	• Selection / execution of each program is independent of each other.
Multiple Tasks (using MASH)	Multiple (≤ 5), but only Single UOP for program execution.	• Multiple programs, e.g. one for each robot, or positioner. • Create a relationship among several individual programs where one is defined as "Head of Family", and others as "Family Member"	• Single UOP to select / execute the Head of Family program, then the entire Family of Programs are selected / executed • Rest of UOPs are for Family member program status monitoring only • Allows for step forward/backward of all programs • Simplifies synchronization between programs

1.3 LIMITATIONS FOR MULTI ARM SYSTEM CONFIGURATIONS

This section describes some of the limitations of standard software features in MultiARM systems with respect to hardware configurations and programming methods described in the "MultiARM Programming Methods" section.

Note You should test each feature described in the following sections in your specific application program to ensure that you understand how each feature works in a MultiARM system. This is especially true in systems that use Type 2 through Type 4 programming methods.

1.3.1 Standard Feature Limitations for MultiARM Systems

This limitation listing applies to standard software features in a MultiARM system. Some of the standard system features do not operate in the same way in MultiARM hardware as they do in single robot systems.

Single Step

Single Step executes motion and logic instructions in teach pendant programs. If a program includes a CALL instruction, the called program will run based on the **step statement type** setting in the Test Cycle SETUP screen. A RUN instruction starts multi-tasking execution, and the multi-tasking

program will single step as well. Any logic controlling or providing signals to peripheral equipment can cause this equipment to move or activate. The following bullets describe Single Step limitations:

- Single step is supported for Type 1 through Type 4 programming. Multi-tasking programs execute lines at the same time, and can end independently. The currently executing line number in each program might be different. BWD execution is applied to multi-tasking programs.

On-The-Fly

The following bullets describe On-The-Fly limitations:

- On-the-Fly speed adjusts only one group, which makes it unsuitable for Coordinated motion programs of all types, and potentially dangerous for Type 4 programs. Refer to [Section 1.2.7](#).
- On-the-Fly speed change is not supported in Type 3 through Type 4 programming. On-the-fly process control only affects one equipment at a time. Saving changes from On-the-Fly adjustments is supported only for the current equipment selected in the On-the-Fly screen. If two to four equipment processes need to be adjusted in this manner, two to four passes will need to be executed. Refer to [Section 1.2.7](#).

Scratch Start (ArcTool only)

Scratch Start may behave differently in multi-robot systems depending on the programming method applied:

- Type 1 and Type 3 programming support Scratch Start. The scratch motion will not be coordinated to a leader frame in Type 3 programming. Do not use ArcStart Synchronization in these cases.
- Scratch Start is supported for all and is applied to all robot arms in a program in Type 2 and Type 4 programming. The scratch motion will not be coordinated to a leader frame in Type 4. ArcStart Synchronization is recommended in these cases but should be tested for the particular situations.
- Scratch and overlap distance are not supported in robot to robot coordinated motion

Program Adjust, Program Shift, Mirror Image Utilities

These utilities are not supported for Multi-Robot (Type 2), Coordinated motion programs (Types 3 and 4), and can cause offset miscalculations for any Touch sense programs.

Tool Offset and Frame Offset Utilities

These utilities are not supported for Coordinated motion programs, or for MultiARM Coordinated motion programs.

- Offset miscalculations will occur for any Touch sense programs.

Return to Path Feature

The Return to Path feature works normally for Type 1 through Type 4 programs. When these programs are paused (for example, by a fault), you can turn the teach pendant to ON, select the motion group you need to jog, then jog that group. Before automatic operation is resumed, jog the group to a safe location, disable the teach pendant, and when the Start signal is input, the motion groups will return to the paused position, and restart the application process. ArcTool returns to the paused position with overlap (if enabled).

Weld Start Synchronization (ArcTool only)

This feature is not supported in Type 1 and Type 2 programming.

- WeldStart Synchronization is not supported across multi-task programs. It is only supported for multiple robots in the same program.
- Alternative methods such as I/O handshaking or Process Synchronization should be used to support *nearly* synchronized weld starts across multi-task programs. The WeldStart Synchronization feature is covered in the *ArcTool Setup and Operations Manual* in the Multi-Equipment section.

Standard Coordinated Motion

Independent Axes and Index device groups cannot be used for coordinated motion leaders or followers. Use robot arms as followers. Basic or General Positioner devices are used as leader groups for Standard Coordinated motion.

Robot-to-robot coordinated motion is supported, but the follower robot cannot be used as a leader robot in any other CD pair definition.

Coordinated Motion Plus

Coordinated Motion Plus allows you to define robot arms as both followers and leaders. Additional CD pair definitions are provided, and a Coord[m] instruction is added to specify the leader group used in a coordinated motion instruction. Toggling Coord[m] to Coord[n] requires a non-coordinated motion (J P[] F/CNT or a L P[] Fine) between the change. Coord[m] will be required in all cases where the coordinated motion leader group cannot be uniquely defined by the program group mask.

Touch Sensing

Touch Sensing has the following considerations:

- Robot-to-robot touch sensing is not supported.
- Standard touch sensing of fixed workpiece(s) is supported.
- Combining offsets from independent touch sense programs for a multi-robot program is not supported.

1.3.2 Special Considerations for MultiARM Systems with Positioner Groups

This section describes limitations for MultiARM systems with positioners or with positioner and index device groups.

Configuration Notes

These configuration are applicable to Ferris wheel, H-frame, and multiple instances of single or multi-axis positioner devices. For ArcTool only, these cell configurations can use motion features and Arc Start Synchronization features. Refer to the ArcTool Setup and Operations manual and the "Multi UOP Interface," and "Multi-Robot I/O Setup" sections of this manual.

Arc Start Synchronization (ArcTool only)

This feature is supported for Type 3 and Type 4 programming.

1.4 HARDWARE OVERVIEW

To use MultiARM, your controller must meet the following basic hardware requirements:

- Your controller must have MultiARM hardware installed.
- Process I/O, ArcLink I/O, DeviceNet, and ArcLink via Ethernet interfaces are supported for welder control. (ArcTool)
- Process equipment as required by the application.
- Interface I/O for a cell controller or remote Operator Panel.

1.4.1 Hardware Definitions

Refer to [Table 1-9](#) for a definition of hardware terminology used in this section.

Table 1–9. Hardware Definitions

TERM	DESCRIPTION
KINEMATICS	Kinematics defines how a robot or positioner will move in Cartesian space (x, y, and z) when the motors (axes) are turned. This is determined by the Cartesian offset information of each axis to the next axis, and the gear ratios. Kinematics is determined for the arm by the robot library that is loaded. Kinematics data are determined for basic and general positioners by the CD_pair calibration.
ROBOT	A motion group entity that is made up of 4 to 6 axes with 1 to 3 optional auxiliary axes. Full forward and backward kinematics are defined for this group. In order for a robot to be used as a follower in Coordinated Motion, the robot must have at least 6 axes (or 6 degrees of freedom).
INDEPENDENT AXIS DEVICE	A motion group entity that is made up of 1 to 3 axes with 1 to 3 optional auxiliary axes. This group does not have coordinated motion, and no kinematics are defined for this group
BASIC POSITIONER	A motion group entity 1 to 3 axes with 1 to 3 optional auxiliary axes. This group can have coordinated motion if the group axes of the positioner are perpendicular (orthogonal axes) or parallel to each other. Only forward kinematics is defined for this group after coordinated motion calibration has been performed.
GENERAL POSITIONER	Group entity 1 to 3 axes, can have 1 to 3 auxiliary axes, can have coordinated motion, group axes of the position do not have to be perpendicular or parallel to each other (non-orthogonal axes), only forward kinematics is defined for this group when coordinated motion calibration is performed.
INDEX DEVICE	Group entity of 1 axis, cannot have coordinated motion, comes with SYSTEM MACROS (1..8) which are defined at setup time (default is 2, 0 degrees, 180 degrees; the user can assign 6 more) which move the axis to a given angle position and set an output. These are protected macro programs that cannot be changed. Further, the user cannot create a program with the index axis group in the group mask. It also supports continuous turn (dial table applications use this feature).  Warning Continuous turn can only be applied to index devices in MultiARM systems. Do not set up continuous turn on the positioner or robot groups because unexpected motion can occur.

Table 1–9. Hardware Definitions (Cont'd)

TERM	DESCRIPTION
Isolation Switch Panel	This panel contains 2 - 4 switches for isolating 1 - 4 robots in a MultiARM system. It is inserted into the master controller door knockout panel.
Fiber optic cable	The fiber optic cable(s) connect the slave controller amplifier(s) to the master controller axis control boards. They send motion control signals and receive encoder and amplifier feedback information in the MultiARM system. Fiber optic cable connections connect amplifiers to the axis control card via the FSSB channels. FSSB-1 is COP10A-1 on the CPU servo axis card. FSSB-2 is COP10A-2 on the CPU servo axis card (COP10A-2 is only present when the CPU servo axis card controls 24 or 36 axes). FSSB-3 is COP10A-3 on the 1st auxilliary axis control card (FSSB-4 is unused). FSSB-5 is on the 2nd auxilliary axis control card. The FSSB can daisy chain through several amplifiers.

1.4.2 System Hardware Setup Information

The following sections outline how the configuration process for MultiARM motion groups and auxiliary axes is performed.

1.4.2.1 FSSB Hardware

- An FSSB channel is the daisy chain connection of axes on amplifiers by fiber optic cables to the axes control DSP devices on the axis control boards. There may be up to 4 FSSB channels defined on the system depending on the axis control boards that are installed.

DSP devices each can control 6 motion axes, so a 12 axes control board will have 2 DSP devices on the board; a 24 axes control board will have 4 DSP devices, etc.

- The main CPU axis control board has FSSB1 and in some cases a second FSSB2 on it, with each FSSB line capable of controlling up to 32 axes (but up to a total of 36 axes for both FSSB1 and FSSB2).

The main CPU axis control board is available in 6, 12, 18, 24, and 36 axes of motion control.

FSSB2 is available only on the 24 and 36 axes main CPU axis control boards.

The amp numbers start with FSSB1 (amps 1 to n) then sequentially to FSSB2, then to FSSB3, and to FSSB5 (note: FSSB4 is unused). The hardware start number for a motion group is the starting DSP axis number for that group. The hardware start number for the first group is "1" on FSSB1, and is automatically assigned. If a dual/triple axis amplifier is used for two/three separate motion groups, then they should be defined as separate amp numbers. A motion group cannot be split across two FSSB's.

- Axes are sequentially assigned to DSP devices; robot 1 (group 1) will use DSP axes 1-6 on DSP1. The next motion axis or motion group on FSSB1 will begin at the 7th DSP axes (on DSP2.) Two six axes robots will require two DSP devices. On a 24 axes board this would leave 12 DSP axes (13-24) available for other non-robot groups on FSSB1 or can be assigned as 12 axes on FSSB2. If we configured robot 2 on FSSB2, it would start at DSP2 DSP axis 7.

The hardware start number (starting DSP axis number) for FSSB2 is dependent on the number of DSP axes used on FSSB1. It must be greater than the number of DSP axes used in FSSB1. Unlike R-30iA, DSP devices can now be split across FSSB channels. That is, remaining axes on the last DSP used on FSSB1 can be used in FSSB2. Therefore, hardware start number for FSSB2 does NOT need to be “multiple of DSP axes (which is 6) + 1”

- FSSB3 is on the 1st auxiliary axis control board which control 24 axes. FSSB3 has a hardware start number (starting DSP number) of 37 — 60. FSSB4 is unused. FSSB5 is on the 2nd auxiliary axis control board which control 24 axes. FSSB5 has a hardware start number (starting DSP number) of 61 — 84.
- Refer to the *Software Installation Manual*, Appendix H for more details and examples regarding FSSB configurations.

1.4.2.2 ServoTorch Harware

(ArcTool) Servotorch is limited to two units.

1.5 SETUP INFORMATION FOR MULTI ARM SOFTWARE AND OPTIONS

MultiARM software uses many standard system features. This section includes special setup information for the standard features.

1.5.1 Installing MultiARM Software

Refer to the *Software Installation Manual* for information about performing a full software load on the robot controller.

Installation automatically loads this software and performs basic robot setup, however, some information concerning the hardware installation of the robots will be requested during installation. Extended axes, extra motion groups, equipment selection, and multi-equipment setup are performed at software installation and require configuration information from the installer. The following sections cover these steps or indicate an appropriate manual to consult for performing these operations.

1.5.2 Setting Up a Positioner or Index Device

Refer to the *Auxiliary Axis Connection and Maintenance Manual* and the *ARC Mate Ferris Wheel 1000 and Weld Module Parts and Service Manual* for more information about how to set up a positioner for your MultiARM system.

1.5.3 Setting Up Multi-Group Motion

Refer to the *Coordinated Motion Setup and Operations Manual* for additional information about setting up and programming multi-group motion.

1.5.4 Setting Up Coordinated Motion

Refer to the *Coordinated Motion Setup and Operations Manual* for detailed information about how to set up coordinated motion for both robot arms. This option is loaded when Coord-Motion Plus Package R669 is installed.

Standard coordinated motion is supported in cases where one robot and one positioner are calibrated, run in the same program, and use COORD motion option on motion statements. Two programs defined in this manner can be executed simultaneously through multi-tasking. The group masks of these programs must be exclusive.

1.5.5 Setting Up Multi-Equipment (ArcTool only)

If you are using multiple weld equipments, you will need to install and configure the Multi-Equipment option. Refer to the Multi-Equipment chapter in the *ArcTool Setup and Operations manual* for information about how to set up and use the Multi-Equipment feature.

1.5.6 Setting Up Multi-Group Weave (ArcTool only)

MultiARM systems support weaving on all robot groups. Weave setup is performed using [Procedure 1-1](#). Weave is only supported on robot groups.

Procedure 1-1 Procedure 5 Setting Up Multi-Group Weaving

Steps

1. Press MENU and select SETUP.
2. Press F1, [TYPE], and select WEAVE. You will see a screen similar to the following.

SETUP Weave		1/12
NAME	VALUE	
1 Weave Enable Groups	[1,*,*,*,*,*,*,*,*]	
2 Dwell delay type:	Move	
3 Frame type:	Tool&Path	
4 Elevation:	0	deg
5 Azimuth:	0	deg
6 Center rise:	0.0	mm
7 Radius:	0.0	mm
8 Blend weave end:	YES	
9 Robot group:	1	
10 Peak output port DO:0		
11 Peak output pulse:	.10	sec
12 Peak output shift:	.08	sec

3. Move the cursor to select Weave Enable Group Mask.
4. Change the second robot group bit from * to 1.
5. Turn off the controller and turn it back on.

This will enable weaving on the second robot. If you want to disable weaving on a robot group, repeat [Procedure 1-1](#) , but in [Step 3](#) , change the robot group bit from 1 to *. Refer to the *ArcTool Setup and Operations Manual* section titled "Multi-Group Weave" for additional detail about setup and use.

1.5.7 Setting Up WeldStart Synchronization for MultiARM (ArcTool only)

To use WeldStart Synchronization the following items need to be set up:

- Configure the system for multi-equipment.
- Set global error ON only if is acceptable or required that all other weld equipment stops welding when any weld equipment is stopped by weld alarms. At controlled start, Press MENU, select Variables, and set \$AWSCFG.\$GLOBAL_ER to TRUE. When this variable is set to FALSE, the operation of one weld equipment does not affect the other equipment.
- Setup Synchronized Weld Schedules. Refer to the Multi-Equipment chapter of the *ArcTool Setup and Operations Manual*, in the Weld Start Synchronization chapter for information on how to synchronize weld schedules.

Note A program using synchronized weld schedules will not execute an WeldStart unless the other equipment is executing a synchronized WeldStart. All weld equipment included in the

synchronized arc weld schedule must set their “Arc Detect” (weld input) within the shortest “Arc detect time” of the equipment included in the synchronized arc weld schedule.

Note Synchronized arc weld schedules will not execute in programs that do not include the equipment indicated in the sync mask, or if an WeldStart for that equipment is not executed. Change the schedule number in the program to a different weld schedule that has the same welding parameters but has not been set up as a synchronized arc schedule.

Note WeldStart instructions that have direct value entries cannot be synchronized. Direct value is not supported for multi-process welders connected via DeviceNet or ArcLink.

1.6 MULTI UOP INTERFACE

The Multi UOP Interface option consists of hardware and software that controls multiple robots with a single controller. Multi UOP Interface adds the necessary control signals to the UOP interface to independently select, start and stop, and monitor independent programs execution on all robot groups. Changes to the operating system monitor these new signals and provide you with the ability control up to five programs independently through addition of UOP signals.

By default, the system has only one set of UOP (Standard UOP). Based on the actual system and application, the actual number of UOPs will need to be configured by the user. Typically, one set of UOP (UOPn/PROGRAMn) could be assigned to each robot. For example, in a QuadARM system, four sets UOPs could be assigned to the four robots. The remaining set of UOP could be assigned to the Positioners.

The Robot Isolation Option allows you to disable motion groups. This feature locks out one or more motion group(s) so that no motion occurs by jogging or program execution. This is especially useful for touching up multi-robot programs. When one robot has been isolated, the other motion groups can be jogged or can be moved by executing the robot program (any positioners included in the program also move). When a position is touched up only the active robot's position data is changed, the position data of the isolated robot group(s) remains unchanged. Refer to [Section 1.7.1](#) for more information on the Robot Isolation feature.

1.6.1 Features and Benefits

A MultiARM system offers the following features and benefits

- **No interlock wiring required between multiple robot controllers** - Independent robots require many input and output signal lines to confirm position between the robots. With MultiARM, the two independent robot programs can interlock easily with internal register, Flags, Marker, I/O, Process Synch option commands.
- **Reduced teaching effort for multiple robots on a single controller system** - All robots are controlled from one teach pendant. Operator does not need to handle multiple teach pendants.

- **Multiple robots share one emergency stop circuit** .- This reduces the wiring between the emergency stop circuits for multiple robots.
- **Fewer unique system backups required** - In these systems all robots are backed up at once where previously each robot was backup up separately.
- **Reduced number of network connections required** A single Ethernet connection replaces the hardware and setup that individual controllers required.
- **Robot Isolation Function** - This feature allows motion groups to be locked out so that program testing can be isolated to only the required motion groups in multi-group programs. For example, all robots except one could be isolated to work on welding parameters for the active robot.
- **UOP Interface Control of Multiple Robots** - Additional signals provide unique control for Program Control:
 - HOLD 2-5, START 2-5, PNSTROBE 2-5, PROD_START 2-5, CYCLE_STOP 2-5
 - RSR9-12, RSR13-16, RSR17-20
 - CMD_ENAB#2-5, SRDY#2-5, HELD#2-5, FAULT#2-5, PROGRUN#2-5, PAUSED#2-5
 - ACK9-12, ACK13-16, ACK17-20

1.6.2 Methods of Robot Program Control

Multi UOP Interface software can support up to five sets of UOPs, with each set controlling one program. e.g. 1st set of UOP controls PROGRAM#1, 2nd set of UOP controls PROGRAM#2, ... etc , until PROGRAM#5. The number of UOP(s)/Program(s) the system supports is determined by the setting of \$MULTI_ROBO.\$NUM_PROG (which is 1 by default). See [Table 1–16](#) for details. In a typical DualARM system, two robot arms are configured as motion group 1 and motion group 2; there may be additional positioner groups installed on these controllers. When the user selects a program, it is designated as program #1 or program #2 based on the program group mask and the default jog group. Program #1 and #2 is indicated in the program select menu next to the program comment. See the following screen for an example.

Select

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No.	Program name	Comment
6	AX_HOME	[]
7	AX_MAIN	[]
8	AX_MOV	[]
9	AX_MW1	#1 []
10	AX_MW1_M	[]
11	AX_MW2	[]
12	AX_MW2_M	[]
13	AX_SW1	#2 []
14	AX_SW1_S	[]

Note The assignment of program #1 or #2 is covered in greater detail in [Section 1.11](#) . Program #1 is executed and monitored by the original UOP signals. Program #2 is executed and monitored by the new UOP control signals that are labeled with #2, such as START#2. When additional robots are present, and the number of UOP program control interfaces could be increased (see [Table 1–16](#) on how to set \$MULTI_ROBO.\$NUM_PROG, and \$MULTI_ROBO.\$PROGC[], and [Table 1–15](#) for the extended UOPs), then Program #3, Program #4, and Program #5 are displayed in the SELECT screen when a program is selected by the respective UOP signals.

The Multi UOP Interface allows the programmer to use different programming structures to accomplish multi-tasking programming. For example, the following types of robot program execution methods can be controlled through Multi UOP Interface.

Table 1–10. Program #1 and Program #2 Control of Program Execution

Program #1	Program #2
Single Robot program execution.	• Single Robot program execution • Single Robot with positioner or index device • Positioner or index device program • Program with no motion control Note: Program #2 group mask must be exclusive of Program #1 group mask for simultaneous execution.
Multi-Robot program, with or without positioner or index device.	• Positioner or index device program • Program with no motion control Note: Program #2 group mask must be exclusive of Program #1 group mask.

Note If your application requires separate programs for each robot to be controlled by separate UOP, set \$MULTI_ROBO.\$NUM_PROG = number of UOPs (≤ 5). If your application require only a single program with multi-groups, a single UOP would be sufficient. Then leave \$MULTI_ROBO.\$NUM_PROG with the default value of one.

1.7 MULTIARM I/O SETUP

MultiARM controllers use the following I/O signals:

- **Digital, Analog, Group, SOP, Link Device, DeviceNet, and others**

The number and types of I/O are dependent on the requirements of the user. Standard installed I/O is covered in [Section 1.8](#), “Modified UOP I/O In a MultiARM System.”

- **Multi-Equipment Interface**

Each process equipment has an interface to the controller. Refer to the chapter “Multi-Equipment” for more information about multi-equipment support.

- **Over Travel/Hand Broken signal and EE Interface**

Each robot has its own OVERTRAVEL, HANDBROKEN , and EE interface. The system detects the number of robot arms installed and provides an ENABLE / DISABLE setting for each group in the MENU/SYSTEM/CONFIGURATION screen.

Robot inputs and outputs run sequentially across the robot arms. Each robot has its own EE connector.

Table 1–11. Robot I/O Mapping

Robot	I/O Assignment
Robot 1	RO[1.. 8] RI[1.. 8]
Robot 2	RO[9..16] RI[9..16]
Robot 3	RO[17..24] RI[17..24]
Robot 4	RO[25..32] RI[25..32]

Note that on the compact EE connector, the Robot I/O start points are as indicated in the above table.

- **Emergency Stop Input**

There is one emergency stop circuit in the robot controller. This circuit is shared across the master and slave controller(s) of a MultiARM system controller.

- **Fence circuit**

There is one fence stop circuit. This circuit is shared across the master and slave controller(s) of a MultiARM controller.

1.7.1 Operator Panel Signals

The MultiARM system is equipped with a standard operator panel (SOP) mounted on the master robot controller. An additional isolation switch panel is installed for supporting Robot Isolation option. The Robot Isolation switch panel can have two, three, or four switches depending on the number of slave controllers and robot groups that need to use the Robot Isolation feature. The switches initially are assigned sequentially to the corresponding motion group; SW 1 is assigned to motion group 1, SW2 to motion group 2, and so on. The assignment of motion groups to a particular switch can be changed or removed. Multiple motion groups can be controlled by one switch.

Refer to [Table 1–12](#) for descriptions of Standard Operator Signals.

Table 1–12. Standard Operator Panel Signals

USE	TYPE OF SIGNAL	DESIGNATION
FAULT POWER	Output	These signals are common to all robots, located on the master controller.
FAULT RESET CYCLE START Auto, T1 / T2 keyswitch	Input	These signals are common to all robots, located on the master controller.
“Robot 1” and “Robot 2”, “Robot 3” and Robot 4 Enable/Disable keyswitches	Input	Each keyswitch enables or disables motion groups.

Refer to [Table 1–13](#) for a description of SOP commands.

Table 1–13. SOP Commands

SOP ACTION	RESULT
FAULT RESET	When this button is pressed, the alarm status is reset for both robots. If servo power is OFF before the Reset input signal, then servo power is turned ON when the alarm is reset.
CYCLE START	When this button is pressed, the selected program starts when the local condition is met.
Enable / Disable keyswitch labeled ROBOT 1, ROBOT 2, ROBOT 3, ROBOT 4	When a robot is disabled from an enabled state, an alarm is posted and the machine lock for that group (robot) is automatically turned ON. When a group (robot) is enabled from disabled status, an alarm is posted and machine-lock is turned OFF automatically. The following alarms are posted for the above Enable/Disable operations. The severity of these alarms is WARN type. • System-029 Robot was connected (G:) • System-030 Robot was isolated (G:) When a keyswitch is set to DISABLED, the motion group(s) that is(are) assigned to that keyswitch are disabled from moving. The value of \$MCR_GRP[<grp num>].\$MACHINELOCK is set to TRUE for this (these) groups. When a motion group is isolated, it will not move by jogging or program execution, however the Position Display of “Isolated” groups will update as though the group was in motion. When training new positions or touching up existing positions that include “Isolated” motion groups, the value for the “Isolated” group will not be changed; in new taught positions the value will be uninitialized and in existing positions, that group’s data will not be changed.

**Warning**

Move the robots to safe position and confirm all the programs are aborted. Otherwise, you could injure personnel, or damage equipment.

1.7.2 Robot Isolation Keyswitch

Two or more Isolation keyswitches are located adjacent to the existing Standard Operator Panel (SOP) switches for Multi-Robot Control. These are provided in a panel and are installed in an existing knockout panel on the controller door. the keyswitches have the following labels (function):

- ROBOT #1 (Enable/Disable) basic unit
- ROBOT #2 (Enable/Disable) basic unit
- ROBOT #3 (Enable/Disable) Option 1 hardware has three isolation switches.
- ROBOT #4 (Enable/Disable) Option 2 hardware has four isolation switches.

The basic hardware signals act on motion groups 1 and 2, which are typically robot #1 and robot #2 in DualARM configurations. With additional arms, it might be desirable to use Option 1 or Option 2 hardware to control more motion groups. For Option 1 and 2, the group assignment is sequential; for example, group 3 to keyswitch 3 and group 4 to keyswitch 4.

Table 1–14.

System Variables	Type	Function	Instructions
\$ROBOT_ISOL	Integer 0 or 1	Value of 1 turns the Robot Isolation function ON or OFF for all affected groups	Set value, then cycle power to take effect
\$ROBOT_ISOLC[#]	Bitmask value	Configures a Keyswitch to control one or more motion groups	Bitmask values for motion groups are (1,2,4,8,16,32,64,128). The sum of the values of the controlled groups is entered. Do not map any group to more than one switch. Use the lowest robot group number when combining multiple groups on a switch (for example, controlling a robot and the positioner that it powers are combined on one switch.) Cycle power to take effect.

Note If your operator panel does not provide the isolation keyswitches, you must turn the Robot Isolation function OFF by setting system variable \$ROBOT_ISOL to 0. Do this at load time at CTRL start, or initiate a CTRL start and then set the variable (see the "Startup Methods" section in the "System Operations" Appendix of your Tool-Specific Setup and Operations manual.)

Notes for setting Robot Isolation operation:

- The initial assignment of groups to keyswitches is switch 1 to group 1 (value =1), switch 2 to group 2 (value =2), switch 3 to group 3 (value =4), switch 4 to group 4 (value =8), \$ROBOT_ISOLC[5] exists but there is no switch 5 (value =16), without respect to what type of group it is.
- The Isolation feature can control individual or multiple groups on each keyswitch.

- Groups cannot be assigned to more than one keyswitch.
- Typically, isolation is performed on robot groups. Positioner groups are mapped to the keyswitch of their corresponding robots because they share MCC control and amplifier power with a robot group.
- Use the keyswitch corresponding to the lowest robot group number when combining control for multiple groups on one keyswitch.
- Assign a value of zero to \$ROBOT_ISOLC[#] when there is no group being controlled by a keyswitch.
- Failure to re-assign a positioner group from a default switch to the same switch as the robot that powers it, will result in a SRVO-117 "Brake Number Conflict" alarm when your controller powers up. (As an example, group 3 positioner is powered by group2 robot. Isolation switch 2 needs to be assigned to control group2 and group3, so \$robo_isolc[2] = 6 (bitmask value group2 = 2 and group3 = 4. Set \$robo_isolc[3] = 0).
- The sum of the bitmask values of the controlled groups is entered in \$ROBOT_ISOLC[#] where “#” refers to a keyswitch. The bitmask values for motion groups are (1,2,4,8,16,32,64,128).
- If a keyswitch is not used (assigned a zero to \$ROBOT_ISOLC[#]), do not DISABLE the keyswitch.
- Failure to assign a positioner group will result in a SRVO-021 "SRDY off (Group:# Axis:*)" alarm when the robot that powers the positioner is isolated.
- It is recommended that you do not change the ENABLE /DISABLE state of a keyswitch when programs are running.
- After you change robot isolation system variables, you must cycle power cycle for the changes to take effect.

Procedure 1-2 Enabling and Disabling the Robot Isolation Feature

Steps

1. Press MENU.
2. Select System.
3. Select Variables.
4. Move the cursor to \$ROBO_ISOL / \$ROBO_ISOLC.
5. Set \$ROBO_ISOL to 1 (ON) or 0 (OFF) (the default is 0 (ON).)
6. Turn controller power off then on again for the change to take effect.

Procedure 1-3 Setting the Isolation Switch Group Mask Values

Steps

1. Press MENU.
2. Select System.

3. Select Variables.
4. Move the cursor to \$ROBO_ISOLC then press ENTER.
5. Move the cursor to \$ROBO_ISOLC[1-4].
6. Set the value of the groups that the switch will control. The value you enter is the sum of the bitmask values of the motion groups to be controlled. The bitmask values for motion groups are:
 - Group 1 = 1
 - Group 2 = 2
 - Group 3 = 4
 - Group 4 = 8
 - Group 5 = 16
 - Group 6 = 32
 - Group 7 = 64
 - Group 8 = 128

Determining the correct value is a two step process:

- a. Determine if the keyswitch default value is associated to a robot group (Group 1 robot is always assigned correctly to switch 1.) If not, determine which slave controller EMG distribution cable is connected to the switch you are setting, set the bitmask value of the robot group to \$ROBO_ISOLC[n].
- b. Next, determine if there are any positioner, index device, independent axis, or special motion group that are powered and controlled by the robot group hardware. Add the bitmask value of that group to \$ROBO_ISOLC[n] from the prior step. Note that a process axis (eg. Servo Torch, Servo dispenser, etc.) is not a motion group.

Examples

Example 1

For group 1 (robot) + group 2 (positioner) on keyswitch 1 Set \$ROBOT_ISOLC[1] = 3 and set \$ROBOT_ISOLC[2] = 0. If there are no other groups, then set \$ROBOT_ISOLC[3-5] = 0

Example 2

For group 2 (robot) + group 3(positioner on robot 2) on keyswitch 2 Set \$ROBOT_ISOLC[2] = 6 and set \$ROBOT_ISOLC[3] =0; \$ROBOT_ISOLC[1] = 1 already. If there are no other groups, then set \$ROBOT_ISOLC[4-5] = 0

Example 3

For group 2 (positioner on robot 1) + group 3 (positioner on robot 1) on keyswitch 1 Set \$ROBOT_ISOLC[1] = 7 and set \$ROBOT_ISOLC[2-3] =0. If there are no other groups, then

set \$ROBOT_ISOLC[4-5] = 0 ex4: For group 2(positioner on robot 1) + group 4(positioner on robot 2) on keyswitch 1 Set \$ROBOT_ISOLC[1] = 3, set \$ROBOT_ISOLC[2] = 12, and set \$ROBOT_ISOLC[3-4] = 0. If there are no other groups, then set \$ROBOT_ISOLC[5] = 0

7. Cycle power for the change to take effect.

1.7.3 Setting the Remote/Local Mode

To set the remote/local mode, use [Procedure 1-4](#) .

Procedure 1-4 Setting the Remote/Local Mode

Steps

1. Press MENU.
2. Select SYSTEM.
3. Select Config.
4. Move the cursor to Remote/Local setup.
5. Press F4, [CHOICE].
6. Select Local or Remote mode according to your requirements.

REMOTE Indicator

The REMOTE indicator is ON when the following conditions are satisfied.

- Remote condition is satisfied (Remote condition is described later)
- No fault condition exists

REMOTE Condition

You can run the system in remote mode (from the user operator panel) when the following conditions are met.

- TP enabled key is OFF
- REMOTE/LOCAL configuration setting item is set to REMOTE.
- UOP *SFSPD signal is ON.
- UOP ENBL input signal is ON.
- System variable \$RMT_MASTER is set to 0 (0=External Device=UOP Panel or PLC)

When the robot is in local mode, if the operator panel condition is satisfied, a robot program can be started by the SOP START button.

1.7.4 User Operator Panel (UOP) Signals

A single arm controller is provided with UOP input and output signals that are used to control single program operation via a remote operator station or programmable logic controller (PLC). MultiARM controllers have Multi UOP Interface software installed, which could be used to configure additional UOP inputs and outputs to control execution of independent programs. Default is one program, however, user may configure the Mutli UOP Interface to provide independent control of up to five programs. [Table 1–15](#) shows the standard and the additional UOP signals.

When you have installed the Multi UOP Interface option (included in the MultiARM Package), the number of available signal resources for normal digital (DI/DO) signals is reduced. The addition of I/O points to the UOP on an existing system may require additional I/O hardware or could require changes to physical wiring such as the pin assignments in CRM2A and CRM2B 50 pin Honda connectors on the process I/O or Terminal I/O blocks. The new I/O points do not have to be mapped sequentially with prior I/O, but may be mapped to higher digital points so that a local "standard" I/O mapping can be retained throughout a site.

Table 1–15. Standard and Extended User Operator Panel I/O signals

SIGNAL TYPE	INPUT	OUTPUT
Standard UOP	UI[1] *IMSTP UI[2] *Hold UI[3] *SFSPD UI[4] Cycle stop UI[5] Fault reset UI[6] Start UI[7] Home UI[8] Enable UI[9] RSR1/PNS1/STYLE1 UI[10] RSR2/PNS2/STYLE2 UI[11] RSR3/PNS3/STYLE3 UI[12] RSR4/PNS4/STYLE4 UI[13] RSR5/PNS5/STYLE5 UI[14] RSR6/PNS6/STYLE6 UI[15] RSR7/PNS7/STYLE7 UI[16] RSR8/PNS8/STYLE8 UI[17] PNS strobe UI[18] Prod start	UO[1] Cmd enabled UO[2] System ready UO[3] Prg running UO[4] Prg paused UO[5] Motion held UO[6] Fault UO[7] At perch UO[8] TP enabled UO[9] Batt alarm UO[10] Busy UO[11] ACK1/SNO1 UO[12] ACK2/SNO2 UO[13] ACK3/SNO3 UO[14] ACK4/SNO4 UO[15] ACK5/SNO5 UO[16] ACK6/SNO6 UO[17] ACK7/SNO7 UO[18] ACK8/SNO8 UO[19] SNACK UO[20] Reserved
Multi Robot Extended UOP (when \$multi_robo.\$num_prog >= 2)	UI[19] *Hold 2 UI[20] Start 2 UI[21] PNS Strobe 2 UI[22] Prod start 2 UI[23] Cycle stop 2	UO[21] Cmd enabled2 UO[22] System rdy 2 UO[23] Motion held2 UO[24] Fault 2 UO[25] Prg running2 UO[26] Prg paused 2
Multi Robot Extended UOP (when \$multi_robo.\$num_prog >= 3)	UI[24] RSR9 UI[25] RSR10 UI[26] RSR11 UI[27] RSR12 UI[28] *Hold 3 UI[29] Start 3 UI[30] PNS strobe 3 UI[31] Prod start 3 UI[32] Cycle stop 3	UO[27] ACK9 UO[28] ACK10 UO[29] ACK11 UO[30] ACK12 UO[31] Cmd enabled3 UO[32] System rdy 3 UO[33] Motion held3 UO[34] Fault 3 UO[35] Prg running3 UO[36] Prg paused 3

Table 1–15. Standard and Extended User Operator Panel I/O signals (Cont'd)

SIGNAL TYPE	INPUT	OUTPUT
Multi Robot Extended UOP (when \$multi_robo.\$num_prog >= 4)	UI[33] RSR13 UI[34] RSR14 UI[35] RSR15 UI[36] RSR16 UI[37] *Hold 4 UI[38] Start 4 UI[39] PNS strobe 4 UI[40] Prod start 4 UI[41] Cycle stop 4	UO[37] ACK13 UO[38] ACK14 UO[39] ACK15 UO[40] ACK16 UO[41] Cmd enabled4 UO[42] System rdy 4 UO[43] Motion held4 UO[44] Fault 4 UO[45] Prg running4 UO[46] Prg paused 4
Multi Robot Extended UOP (when \$multi_robo.\$num_prog >= 5)	UI[42] RSR17 UI[43] RSR18 UI[44] RSR19 UI[45] RSR20 UI[46] *Hold 5 UI[47] Start 5 UI[48] PNS strobe 5 UI[49] Prod start 5 UI[50] Cycle stop 5	UO[47] ACK17 UO[48] ACK18 UO[49] ACK19 UO[50] ACK20 UO[51] Cmd enabled5 UO[52] System rdy 5 UO[53] Motion held5 UO[54] Fault 5 UO[55] Prg running5 UO[56] Prg paused 5
COMPARISON OF SINGLE TO MULTI-ROBOT USAGE	Total 50 points (In single robot system, there are 18 UOP's)	Total 56 points (In single robot system, there are 20 UOP's)

1.8 MULTI UOP INTERFACE: EXTENDED UOP

Multi UOP Interface software provides a modified UOP for independent robot control. The I/O signal assignments can be reviewed by following [Procedure 1-5](#) .

Procedure 1-5 Displaying MultiARM System I/O**Steps**

1. Press MENU, and select I/O.
2. Press F1, [TYPE], to display I/O types.
3. Select from weld, digital, analog, group, and so forth.
4. Press the CONFIG key to review the signal mapping.

1.8.1 Setting Up Multi UOP Interface

Multi UOP Interface uses the following system variables to set the number of UOP interfaces for program control, and to configure for local or global operating modes.

Multi UOP Interface System Variables

Multi UOP Interface installs system variables that allow users to configure the number of UOPs, and to define which motion groups are intended to be assigned to which UOP#n / PROGRAM#n. The following variables are used for Mutli-Robot control for selection and program execution: \$MULTI_ROBO_T

It is important to setup

- \$MULTI_ROBO.\$NUM_PROG
- \$MULTI_ROBO.\$PROGC[]

properly before programming**Table 1–16. Multi UOP Interface System Variables**

VARIABLE	TYPE	FUNCTION	INSTRUCTION
\$MLT_ENABLE	BOOLEAN	Turns Multi UOP Interface ON. The default is TRUE.	Do NOT change.
\$NUM_PROG	INTEGER	This indicates the number of UOP interfaces that will made for the system.	When the value is 1, standard UOP signals are available. With the value set to 2-5, additional sets of UOP signals will be created. COLD START is required.

Table 1-16. Multi UOP Interface System Variables (Cont'd)

VARIABLE	TYPE	FUNCTION	INSTRUCTION
\$MULTI_PROGC[]	Array[5] of INTEGER	Bitmask value of motion groups permitted to execute as Program#n, controlled by UOP #n signals.	These variables determine which motion groups can be executed by UOP#n signals. The value of is the sum of the group bit mask values of motion groups allowed to execute on that set of UOP controls. If a program has any group# that is contained in the value \$MULTI_PROGC[n], then that program can be selected and executed by UOP#n signals. The defaults are 1, 2, 4, 8, 16 for \$MULTI_PROGC[1] ... \$MULTI_PROGC[5] respectively. The default setting would force "group1 only" programs to use UOP1 signals, "group2 only" programs to use UOP2, etc. Programs that had motion group1 and group2 could execute on UOP1 or UOP2. Setting the value of \$MULTI_PROGC[n] = 7 means that programs containing group 1 or group 2, or group 3 can be executed by UOP "n" signals. Programs with no motion group control can be executed on any UOPn; \$MULTI_PROGC[n] = -1 forces execution of programs that do not control motion to UOP signal "n." \$MULTI_PROGC[n] = 0 is not used. COLD START is required.

Table 1–16. Multi UOP Interface System Variables (Cont'd)

VARIABLE	TYPE	FUNCTION	INSTRUCTION
\$HOLD_TYPE	INTEGER	Configures how HOLD(#) operates. The default is 0, all HOLD.	Set \$HOLD_TYPE = 1, then HOLD(#) signals are Independent; HOLD# holds only program#. Set \$HOLD_TYPE = 0, then HOLD(#) are global; all Program# HOLD when any HOLD(#) signal is turned OFF. Note, HOLD(#) is a normally high signal.
\$D_ARM_LCK	INTEGER	This variable enables and sets control group mask for the DualARM Lock jogging feature. The default is 0, that turns the feature OFF.	DualARM Lock jogging feature is used when Multiple robots are used to move a single object And prevents jogging groups indepedently. \$D_ARM_LCK is a number greater than 2 that is the sum of the bitmask values of all of the motion groups required to jog simultaneously via coordinated motion. No other single group can jog when \$D_ARM_LCK is greater than 0. This variable can be set in a TP PARAMETER instruction.

Table 1–16. Multi UOP Interface System Variables (Cont'd)

VARIABLE	TYPE	FUNCTION	INSTRUCTION
\$TP_DEFPROG1	STRING	(For internal use. Do NOT change its value) This is the name of the program that will execute with “UOP” signals (original single robot control.) (The following applies to all 5 of these variables: The initial value is "uninit"; the value is changed by the program select method. Do not attempt to change these variables manually or programmatically or the system will not work correctly or/and may execute the wrong program that could damage equipment or harm an operator.)	When an “appropriate program” is selected, “#1” will appear next to that program in the Select menu. The “appropriate program” Must contain a motion group in the group mask defined in \$MULTI_PROGC[1] (see below.)
\$TP_DEFPROG2	STRING	(For internal use. Do NOT change its value) This is the name of the program that will execute with “UOP 2” signals	When an “appropriate program” is selected, “#2” will appear next to that program in the Select menu. The “appropriate program” Must contain a motion group in the group mask defined in \$MULTI_PROGC[2] (see below.)
\$TP_DEFPROG3	STRING	(For internal use. Do NOT change its value) This is the name of the program that will execute with “UOP 3” signals.	When an “appropriate program” is selected, “#3” will appear next to that program in the Select menu. The “appropriate program” Must contain a motion group in the group mask defined in \$MULTI_PROGC[3] (see below.)

Table 1–16. Multi UOP Interface System Variables (Cont'd)

VARIABLE	TYPE	FUNCTION	INSTRUCTION
\$TP_DEFPROG4	STRING	(For internal use. Do NOT change its value) This is the name of the program that will execute with “UOP 4” signals.	When an “appropriate program” is selected, “#4” will appear next to that program in the Select menu. The “appropriate program” Must contain a motion group in the group mask defined in \$MULTI_PROGC[4] (see below.)
\$TP_DEFPROG5	STRING	(For internal use. Do NOT change its value) This is the name of the program that will execute with “UOP 5” signals.	When an “appropriate program” is selected, “#5” will appear next to that program in the Select menu. The “appropriate program” Must contain a motion group in the group mask defined in \$MULTI_PROGC[4] (see below.)

Note On prior hardware platforms it was required to manually set the system Variable \$SCR.\$ITP_TIME to a higher setting. MultiARM software makes this setting automatically when the system is loaded.

- When the option is loaded and the variable \$multi_robo \$enable = TRUE, UOPs are reconfigured to support independent control of multiple programs.
- The number of control interfaces can be increased to support up to five independently executing programs by setting the variable \$multi_robo.\$num_prog = 3..5, then cycle power.

1.8.2 Hold Input Configuration

The MultiARM System has up to five UOP HOLD signals (HOLD 2 ... HOLD 5). The function “scope of the “HOLD” signal is determined by setting the following system variable:

- **\$MULTI_ROBO.\$HOLD_TYPE = 0** - This is the default setting that is made when you install the Multi-Robot Option. If any of the HOLD signals are OFF, all the executing programs are paused.
- **\$MULTI_ROBO.\$HOLD_TYPE = 1** - If HOLD is OFF, “PROGRAM#1” is paused. If HOLD2 is OFF, “PROGRAM#2” is paused, etc. for each installed UOP interface. If there are additional executing programs, that are not assigned as PROGRAM#1-Program#5 (eg.

non-motion programs, typically), they cannot be HELD by any HOLD signal. In order to pause these programs, turn off the ENBL input.

Refer to [Section 1.11.1](#) for more details about selection PROGRAM#1 and PROGRAM#5.

Note HOLD signals are normally ON signals.

1.8.3 Start Input Configuration

A Multi-Robot System should be set up so that the UOP Start signal(s) will be START for CONTINUE only. This can be done by setting the system variable \$SHELL_CFG.\$CONT_ONLY to TRUE or using the system configuration menu. The setup is done using [Procedure 1-6](#) .

Procedure 1-6 Setting “START for CONTINUE Only”

Steps

1. Press MENU and select SYSTEM.
2. Press F1, [TYPE], and select Config.
3. Move the cursor to “START for CONTINUE only:”
4. Set the value to TRUE.

When the setting is TRUE, only paused PROGRAM#n can be started by a START #n signal under this setting. In case of starting an aborted program, use the RSR assigned to Program# or PROD START # for PSN signals.

A paused program is restarted from the suspended line number in case that a START# signal is input. The START# signal must correspond to the PROGRAM# for that suspended program to restart. An aborted program, can not be restarted by inputting either START or START#, and the message “PROG-023 Task is not paused” is displayed on the TP error line.

A Multi-Robot System has up to five START signals (START, START 2, START 3, START 4, and START 5).

- START - This start signal is for Program #1.
- START# - This is the start signal for Program #.

1.8.4 RSR Input

Multi-Robot System uses RSR1-RSR4 for PROGRAM#1 and RSR5-RSR8 for PROGRAM#2 and has two queues corresponded to RSR1-4 and RSR5-8. When additional UOP interfaces greater than

2 are configured, each set of control signals includes four more RSR signals (see Table 1-15) and a program queue. RSR signals use the new inputs; PNS does not use these additional signals. The following describes how the RSR signals operate:

- To select and execute an RSR program using UOP interface "n", the motion groups of the RSR program must match the data in \$multi_robo.\$multi_progc[n]. See [Section 1.11.1](#) for additional details.

1.8.5 PNS STROBE# and PROD START# Input

The Multi-Robot System can have up to five PNS STROBE# and PROD START# signals.

- Regardless of the UOP configuration, all PNS STROBE input signals use the value set in UOP UI[9] .. UI[16].
 UI[9] RSR1/PNS1/STYLE1
 UI[10] RSR2/PNS2/STYLE2
 UI[11] RSR3/PNS3/STYLE3
 UI[12] RSR4/PNS4/STYLE4
 UI[13] RSR5/PNS5/STYLE5
 UI[14] RSR6/PNS6/STYLE6
 UI[15] RSR7/PNS7/STYLE7
 UI[16] RSR8/PNS8/STYLE8
- PNS does not use the additional inputs for RSR when the number of UPO interfaces is three or greater.
- To select and execute a PNS program using UOP interface "n", the motion groups of the PNS program must match the data in \$multi_robo.\$multi_progc[n]. See [Section 1.11.1](#) for additional details.

Refer to the “Production Operation Setup” section of the *General Setup* chapter of your application-specific *Setup and Operations Manual* for additional instructions on using the PNS function.

1.8.6 Cmd Enable Input

Multi-Robot Systems have up to five CMD ENABLED signals, Cmd Enabled, Cmd Enabled2.. Cmd Enabled5, that are used for control of Program#1 .. Program#5.

Cmd Enabled is ON if all of the following conditions are satisfied.

- Remote condition is satisfied.
- FAULT, FAULT 2 — 5 are all OFF. (no alarm)

- You do not have the teach pendant in the STEP mode.
- All robots / groups which are configured for PROGRAM1 (See \$MULTI_PROGC[1]) are connected.

Similar conditions apply to CMD ENABLED #2, 3, 4, 5.

To run the system in the Remote Condition:

- The Teach Pendant enable switch is set OFF.
- The system/config local/remote setup is remote.
- The *SFSPD input is ON.
- The ENBL input is ON.
- The system variable \$RMT_MASTER is 0 (UOP= master).

1.8.7 SYSRDY Output

Multi-Robot System has up to five System Ready signals, System Rdy 2..System Rdy 5 that are used for control of Program#1 .. Program#5.

System Ready, System Rdy # are ON if all the following conditions are satisfied:

- All the axes are ready.
- Robot Isolation is Connected for all groups required by the program (see section Robot Isolation for more details.)

This is the same for option 1, three isolation switches, and for option 2, four isolation switch panels.

1.8.8 ATPERCH Output

The specified signal AT Perch is set to ON when the group 1 robot is at reference position 1. ATPERCH functionality is not supplied for additional motion groups. This feature can be had indirectly, however. Each group supports at least 1 reference position which can be assigned to a DO[]. You can assign a reference position for each group to indicate an "at home" status. Index devices provide this feature through setup. Each index position (up to eight total) can have a digital output assigned to turn ON when the specified position is reached. Refer to the "Reference Position Utility" section of the *Advanced Functions* chapter of your application-specific *Setup and Operations Manual* for more information about the utility.

1.8.9 PRG RUNNING and PRG PAUSED Outputs

In a Multi-Robot system, Prg Running and Prg Running2.. Prg_running5 signals are supplied to indicate if Program#1 .. Program#5 is running. Prg Paused and Prg Paused2.. Prg Paused5 indicate if Program#1 .. Program #5 is paused. Additional programs to Program#1 and Program#2 ... Program#5 can be executing in the system via multi-tasking. These programs cannot be monitored via Progran# and Paused# outputs. Program logic must provide a signal if the PLC is to monitor operation of programs that are not Program#1 .. Program#5.

1.8.10 MOTION HELD Output

Multi-Robot Control System has up to five Motion Held signals, Motion Held and Motion Held 2 .. Motion Held 5. Held or Held 2 .. Held 5 is ON if one of the following conditions is satisfied:

- UOP Motion Held or Motion Held 2 .. Motion Held 5 is OFF.
- The HOLD switch on the Teach Pendant is pressed.

Note The normal state for HOLD and HOLD2 is ON.

1.8.11 FAULT Output

Multi-Robot Control System has up to five Fault signals, Fault and Fault 2 .. Fault 5. These indicate the status of Program#1 .. Program#5.

1.8.12 Robot I/O

A Multi-Robot System has ROBOT I/O, Robot Overtravel/Handbreak inputs, and PPABN inputs for each robot.

1.8.13 Making I/O Assignments

Ordinarily, the I/O for equipment and UOP interface are automatically assigned when the software is installed. If you change equipment, it might be necessary to reinitialize the I/O assignment.

Process I/O

Use [Procedure 1-7](#) to make Multi-Robot I/O assignments.

Procedure 1-7 Making Process I/O Assignments Automatically

Conditions

- You have installed the Multi-equipment option.
- You have two equipments controlled by discrete digital and analog I/O.



Caution

This procedure will unassign all I/O. You should determine whether manual assignment of weld I/O should be performed instead of this procedure.

Steps

1. Press MENU, and select I/O.
2. Press F1, [TYPE], and select Link Device.
3. Press F5, CLR_ASG, and Press F4, YES to clear the current I/O assignments.
4. Turn the controller off, and then on. The I/O assignments will automatically be made.

Note Refer to Chapter 3 for information about multiple-equipment setup.

1.9 MULTI ARM SYSTEM MASTERING

You need to perform mastering and calibration on all robots and all additional motion groups in your MultiARM system before you can use it. Mastering a MultiARM system involves the following steps.

- Master each robot (It is recommended that you use the *iRCalibration Vision Mastering* and *iRCalibration Vision TCP Set Options*. Refer to the *iRCalibration Operator Option Manual* for more information.)
- Master each basic positioner, general positioner, independent axis, and index device
- Calibrate each robot-to-robot pair (It is recommended that you use the *iRCalibration Vision Multi-Cal Option*. Refer to the *iRCalibration Operator Option Manual* for more information). Please note that the *iRCalibration Vision Multi-Cal Option* is included in the *Coord-Motion Plus Package*.
- Calibrate each positioner-to-robot pair (It is recommended that you use the *iRCalibration Vision Multi-Cal Option*.)
- Coordinated Motion Setup. Refer to the *Coordinated Motion Setup and Operations Manual*.

1.10 MULTI ARM CALIBRATION

1.10.1 Overview

The iRCalibration Vision Multi-Cal Option can be used to determine the frame relationship between two robots or the frame relationship between a robot and a positioner. For Coordinated Motion, iRCalibration Vision Multi-Cal Option determines the frame of the coordinated pair. iRCalibration Vision Multi-Cal Option is included with the Coord-Motion Plus package (R669). Please refer to the iRCalibration Operator Option manual for more information on its operations.

1.11 MULTI-TASKS EXECUTION BY MULTIPLE UOPs USING MULTI UOP INTERFACE

When the Multi UOP Interface (included in MultiARM Package) is installed, up to five programs can be controlled. You can select programs from the teach pendant, or automated methods can select them through the UOP interface. The program that is selected by UOP interface 1 is referred to as PROGRAM#1, and the program that is selected by UOP interface 2 is referred to as PROGRAM#2. Programs with motion groups other than 1 or 2 can be assigned to PROGRAM#1 or PROGRAM#2.

Multi UOP Interface adds I/O to the system for automatic execution using multitasking. The Multi UOP Interface can be setup to provide independent control of up to five programs. In the following sections, an example DualArm cell designs with robot groups 1 and 2 that operate independently from each other is used. An extension to this configuration is adding positioners that perform dedicated operations with the two robots. In the following descriptions, two sets of UOPs are used.

When programming a MultiARM system, you must train programs and select and execute programs. It requires selecting motion groups. You select which robot to program through the teach pendant. In R-30iB, you could use the “GROUP” key on the iPendant to cycle through the groups and quickly get to the group you want to select. To toggle between motion groups using traditional methods, also see [Procedure 1-8](#) and [Procedure 1-9](#).

Procedure 1-8 Selecting a Robot's Motion Group from the FCTN Menu

Steps

1. Press the FCTN key to display the function menu. You will see a screen similar to the following.

No.	Program	FUNCTIONS	72
		1 ABORT	
20	MAIN	2 Disable FWD BWD	]
21	MAIN2	3 CHANGE GROUP	]
22	MAIN2A	4	]
23	P124	5	]
24	P124TW	6 CHANGE EQUIPMENT	]
25	P12_OUT	7 RELEASE WAIT	]
26	P14	8 TOGGLE COORD JOG	]
27	P14_TWIN	9	]
28	P1A_TWIN	10 -- NEXT --	]
		+-----+	

2. Every time you select CHANGE GROUP, the motion group will be changed to the next group.

For example, in a system with three motion groups, the group will change from group1 to group2, group2 to group3, and group3 to group1 if you select CHANGE GROUP three times.

Procedure 1-9 Changing Motion Groups from the Jog Menu

Steps

1. Press SHIFT and COORD to display the jog menu.
2. Set the cursor to "Group" in the Jog menu and press a number key. Motion group is changed to the one you choose. The currently selected motion group is displayed on the teach pendant status line as "G1,""G2,"and so forth.

	974678		
No.	Program name	Tool (.=10)	1
20	MAIN	Jog	0
21	MAIN2	User	0
22	MAIN2A	Group	2
23	P124	+-----+	
24	P124TW	#2 [	]
25	P12_OUT	[	]
26	P14	#1 [	]
27	P14_TWIN	[	]
28	P1A_TWIN	[	]

Note When PROGRAM#1, PROGRAM#2, PROGRAM#3 etc. have been assigned, changing the jog group from 1 to 2 will toggle between PROGRAM#1 and PROGRAM#2 in the editor automatically, when PROGRAM#1 includes robot 1, but does not include robot 2 and when PROGRAM#2

includes robot 2, but not robot 1 and the system variable \$SCR.\$MAXNUMTASKS = 2. When \$SCR.\$MAXNUMTASKS > 2, the edited program is not changed automatically.

The value of \$SCR.\$MAXNUMTASKS impacts automatic selection and execution as well and is further covered in [Section 1.11.6](#) through [Section 1.11.8](#).

1.11.1 Selecting and Executing PROGRAM#1 and PROGRAM#2

Multi UOP Interface (included in MultiARM Package) provides the means to execute and monitor independent programs simultaneously. For example, two sets of UOPs could be setup for a DualARM system. The Program that is selected for group 1 is referred as PROGRAM#1 and the Program that is selected for group 2 is referred as PROGRAM#2. Program#1 and Program#2 are also assigned to multi-robot motion programs, robot + positioner motion programs, and programs without motion groups. There are manual and automatic selection methods supported.

- Manual method is via the teach pendant SELECT screen
- Automatic method 1, RSR
- Automatic method 2, PNS
- Style Select function does not support multiple “select / execute” operations. The MASH programming method can be used to start multiple programs via the Style Select method.

Table 1-17.

System Variable	Type		
\$MULTI_PROGC	Array[5] of INTEGER	Bitmask value of motion groups permitted to execute as Program#n, controlled by UOP #n signals.	These variables determine what motion groups can be executed by UOP#n signals. The value of is the sum of the group bit mask values of motion groups allowed to execute on that set of UOP controls. If a program has any group# that is contained in the value \$MULTI_PROGC[n], then that program can be selected and executed by UOP#n signals. The defaults are 1, 2, 4, 8, 16 for \$MULTI_PROGC[1] ... \$MULTI_PROGC[5] respectively. The default setting would force "group1 only" programs to use UOP1 signals, "group2 only" programs to use UOP2, etc. Programs that had motion group1 and group2 could execute on UOP1 or UOP2 Setting the value of \$MULTI_PROGC[n] = 7 means that programs containing group 1 or group 2, or group 3 can be executed by UOP "n" signals. Programs with no motion group control can be executed on any UOPn; \$MULTI_PROGC[n] = -1 forces execution of programs that do not control motion to UOP signal "n." \$MULTI_PROGC[n] = 0 is not used

Manual Selection and Execution of Program#1 and Program#2

When you move the cursor to a program in the SELECT screen and press the ENTER key, that program will become the "selected program" and \$TP_DEFPROG is set to this program name. This program is now available for editing and execution via the teach pendant, SOP, or UOP. At the same time, the program is also assigned as PROGRAM#1 or PROGRAM#2 automatically, according to the motion groups in the program's group mask, the jog group indicated in the status line of the teach pendant, and the status of the current PROGRAM#1 and PROGRAM#2.

Table 1–18 contains information about how the group mask in your programs causes them to be set as either PROGRAM#1 or PROGRAM#2 when you select programs.

Table 1–18. Program Group Mask Setting

PROGRAM GROUP MASK	ASSIGNMENT
(1,*,1/*,1/*,1/*)	Will be assigned as PROGRAM#1, the jog group has no influence.
(*,1,1/*,1/*,1/*)	Will be assigned as PROGRAM#2, the jog group has no influence.
(1,1,1/*,1/*,1/*)	Will be assigned as PROGRAM#1, when jog group is group one.
(*,*,1/*,1/*,1/*)	Will be assigned as PROGRAM#1, when jog group is group one.
(1,1,1/*,1/*,1/*)	Will be assigned as PROGRAM#2, when jog group is group two.
(*,*,1/*,1/*,1/*)	Will be assigned as PROGRAM#2, when jog group is group two.
(*,*,1/*,1/*,1/*)	Will assigned as PROGRAM #2 when jog group is group three, four, or five.

Note In Table 1–18, 1/* in the field of the motion group mask specifies that the motion group might not be included in the program. For any assignment to occur, PROGRAM#n cannot be running or paused.

Procedure 1-10 Selecting PROGRAM#1 and PROGRAM#2

Steps

1. Press MENU and choose Select.
2. Select a program from the SELECT screen. It will be assigned as either PROGRAM#1 or PROGRAM#2.

In lines 24 and 26 of the screen below, "#1" is displayed to the right of program name that is used as "PROGRAM#1", and "#2" is displayed to the right of program name which is used as "PROGRAM#2".

- If a paused program (other than the selected program) has a motion group that is required by the selected program, then the selected program can not start.

1.11.3 Simultaneously Running Two Programs from the Teach Pendant

Both PROGRAM#1 and PROGRAM#2 can run at the same time using the multi-tasking control function. Start by selecting the teach pendant programs that are to be PROGRAM#1 and PROGRAM#2.

Procedure 1-11 Running Two Programs from the Teach Pendant

Steps

1. Press MENU and then choose Select.
2. Select the teach pendant programs that are to be PROGRAM#1 and PROGRAM#2
3. Select and set PROGRAM#1 and PROGRAM#2.
4. Set the teach pendant jog group to any group controlled by the program you want to execute first.

The selected program for this group becomes \$TP_DEFPROG, it replaces the current program in the teach pendant editor screen. With the teach pendant enable switch turned ON and the DEADMAN switch ON, you can start the selected program by pressing the SHIFT+FWD buttons.

5. To execute the second program while the first program is executing, change the jog group by using the FCTN menu/CHANGE GROUP, or by using the JOG menu to change the jog group. This program will become the selected program and replaces the current (executing) program in the teach pendant editor.

Pressing SHIFT+FWD buttons will now start execution of the second program if the second program does not use motion groups already executing in the first program.

You must have the jog group equal to one group of the program you want to execute from the teach pendant, otherwise the error "TPIF-044 Program is unsuitable for robot" will be posted.

1.11.4 Running a Program from the SOP

If you want to run a single program from the SOP, use [Procedure 1-12](#) .

Procedure 1-12 Running a Program from the SOP

Steps

1. Select the program you want to run.

2. Turn the teach pendant enable switch OFF.
3. Set the REMOTE/LOCAL to LOCAL on the config menu, and press the Cycle Start button on the SOP.
4. The selected program will start executing.

Note Unlike teach pendant execution of a program, the selected program starts regardless of the jog motion group indicated in the teach pendant status line.

1.11.5 Simultaneously Running Two Program from the SOP

Both PROGRAM#1 and PROGRAM#2 can run at the same time using the multi-tasking control function from the SOP Cycle Start. .

Procedure 1-13 Simultaneously Running Two Programs from the SOP

Steps

1. Select the PROGRAM#1 and PROGRAM#2 you want to run.
2. Turn the teach pendant enable switch OFF.
3. Set the REMOTE/LOCAL config item on the Config menu to LOCAL.
4. Select the first program that you want to execute from the teach pendant SELECT menu.
5. Press Cycle Start on the SOP.
6. Select the second program you want to run. The second program will replace the first selected program in the teach pendant editor.
7. Press Cycle Start on the SOP. Execution of the second program will start if the second program does not use motion groups already executing in the first program. The second program will become the currently selected program.

Note The program you selected in [Step 4](#) will start executing, regardless of the jog group that is currently displayed in the teach pendant status line. The second program selected in [Step 6](#) will execute regardless of the current jog group if there is no group mask conflict with the first program.

1.11.6 RSR Selection and Execution of PROGRAM#1 and PROGRAM#2

You can select programs to run with the RSR or PNS Option. If you are using PNS to select programs, RSR will not be available.

In single robot system:

- When all programs are aborted and RSR1-8 is turned on, the program corresponded to the RSR number is started.

Note If Multi UOP Interface is not installed or is turned off, up to 7 programs can be queued to execute.

- If a program is currently executing and the same RSR number is selected, the input has no effect.
- If the RSR input is not the same as the current program, the request is joined to the queue, and this program executes when the current program completes executing.

In a Multi-Robot system, RSR1-8 is divided to RSR1-4 for Robot#1 and RSR5-8 for Robot#2. There are two queues, one for RSR1-4 and one for RSR5-8 so that a different RSR1-4 and a different RSR5-8 can be queued to run when the respective program finishes executing. The group masks of the programs must meet the requirements for PROGRAM#1 and PROGRAM#2 assignment. RSR1-4 programs must have group1 if they also have group2. RSR5-8 programs must have group2 if they also have group1. Each of the two queues will support up to three programs.

RSR Setup				1/11
1	RSR1	program number	[ENABLE]	[0]
2	RSR2	program number	[ENABLE]	[0]
3	RSR3	program number	[ENABLE]	[0]
4	RSR4	program number	[ENABLE]	[0]
5	RSR5	program number	[DISABLE]	[0]
6	RSR6	program number	[DISABLE]	[0]
7	RSR7	program number	[DISABLE]	[0]
8	RSR8	program number	[DISABLE]	[0]
9	Base number			[0]
10	Acknowledge function		[FALSE]	
11	Acknowledge pulse width(msec)		[400]	

Note In the screen above,

- The program started by RSR1-4 becomes "PROGRAM#1".
- The program started by RSR5-8 becomes "PROGRAM#2".
- The most recently selected program is assigned to \$TP_DEFPROG and can be executed by the teach pendant.

Using RSR to Select and Execute PROGRAM#1 and PROGRAM#2 and to Start Execution

This method provides external selection through UOP inputs. Refer to the “Production Operation” section of the *Testing a Program and Running Production* chapter in your application-specific *Setup and Operations Manual* for setup details for RSR.

Below is the RSR input designation when Multi-Robot Control is loaded.

Table 1–19. Multi-Robot RSR Inputs

RSR SETUP ITEM	DESCRIPTION
RSR1 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR1 signal is received. For example, if you entered 0023 for the RSR1 program number and the base number was set to 100, the RSR1 signal would execute program RSR0123. This program should have group 1 in its group mask. If you enter an invalid program number, the system will ignore this signal.
RSR2 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR2 signal is received. If you enter an invalid program number or a zero, the system will ignore this signal.
RSR3 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR3 signal is received. If you enter an invalid program number or a zero, the system will ignore this signal.
RSR4 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR4 signal is received. If you enter an invalid program number or a zero, the system will ignore this signal.
RSR5 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR5 signal is received. Programs assigned to RSR5-RSR8 should contain group 2 in its motion group mask otherwise it will not execute. If you enter an invalid program number or a zero, the system will ignore this signal.
RSR6 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR6 signal is received. If you enter an invalid program number or a zero, the system will ignore this signal.
RSR7 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR7 signal is received. If you enter an invalid program number or a zero, the system will ignore this signal.
RSR8 Program Number	This item allows you to enter a number that when added to the base number defines the program number that will be executed when the RSR8 signal is received. If you enter an invalid program number or a zero, the system will ignore this signal.
Base Number	This item allows you to enter a number that when added to the RSR1-8 program number defines which program will be executed. This base number can be changed from within your program by using the PARAMETER NAME instruction. The parameter that contains the RSR base number is \$SHELL_CFG.\$JOB_BASE . By changing the base number, you can control which group of programs will be executed.
Acknowledge Function	This item allows you to enable or disable robot acknowledge output signals ACK1-8. FALSE means the signals are disabled. TRUE means the signals are enabled.
Acknowledge Pulse Width	This item allows you to set the length (in milliseconds) of the ACK1-8 signal when the acknowledge function is enabled. This time depends on the scan time of your PLC program.

When Multi-Robot Control is loaded, the setup screen for RSR provides assignment for RSR5 to RSR8, as shown below. The programs that are assigned to RSR5 to RSR8 should contain group two in their group mask.

RSR Setup				1/11
1	RSR1	program number	[ENABLE]	[0]
2	RSR2	program number	[ENABLE]	[0]
3	RSR3	program number	[ENABLE]	[0]
4	RSR4	program number	[ENABLE]	[0]
5	RSR5	program number	[DISABLE]	[0]
6	RSR6	program number	[DISABLE]	[0]
7	RSR7	program number	[DISABLE]	[0]
8	RSR8	program number	[DISABLE]	[0]
9	Base number			[0]
10	Acknowledge function		[FALSE]	
11	Acknowledge pulse width(msec)		[400]	

The user assigns RSR programs to RSR1 to RSR4 and for RSR5 to RSR8. To select these programs, the appropriate RSR input is turned ON. This also causes the RSR program to execute, or be queued for execution.

There are some problems that could prevent RSR from starting your program. If your program will not run, it could be for one of the following reasons:

- The program started by RSR1-4 does not have motion group 1 and it has motion group 2.
- The program started by RSR5-8 does not have motion group 2 and it has motion group 1.
- If the program is currently executing it will not be buffered for future execution.

Note If the group mask of an RSR program contains neither group 1 or group 2, then it can be assigned to RSR1-8.

The three conditions above will cause the following error to be displayed above the status line of the teach pendant.

TPIF-44 Program is unsuitable for robot

If PROGRAM#1 or PROGRAM#2 is executing and an RSR program that is unsuitable to be run from the select menu, it will not be queued, and no error messages will be displayed.

Note When executing, the outputs PROGRUN#1/PAUSED#1 and PROGRUN#2/PAUSE#2 indicate the status of the two programs. In the event either program is paused, the START#1 or START#2 are used to resume operation of PROGRAM#1 and PROGRAM#2 respectively.

1.11.7 Using PNS to Select and Execute PROGRAM#1 and PROGRAM#2

You can select RSR or PNS(Optional). If PNS is selected, RSR is not available. In a DualARM system, there are UOP signals PNSTROBE#1, PNSTROBE#2, PROD_START#1 and PROD_START#2 for controlling program execution. The program selected by PNSTROBE#1 becomes "PROGRAM#1". This program can be started by PROD_START#1 and it is resumed from pause by START#1. The program selected by PNSTROBE#2 becomes "PROGRAM#2". This program can be started by PROD_START#2 and it is resumed from pause by START#2.

In the following cases, the program is not selected.

- The program selected by PNSTROBE#1 does not have motion group 1 and it has motion group 2.
- The program selected by PNSTROBE#2 does not have motion group 2 and it has motion group 1.

When this happens, you will see the following message:

TPIF-44 Program is unsuitable for robot

Recommended Motion Group Settings for PROGRAM#1 and PROGRAM#2

The following are recommended motion groups of the selected program.

- PROGRAM#1 has motion group 1 only
- PROGRAM#2 has motion group 2 only

-OR-

- PROGRAM#1 has motion group 1 and 2.
- PROGRAM#2 does not have any motion group, or only groups 3, 4, 5.

Note If the group mask of a PNS program contains neither group 1 or group 2, it can be selected and executed as either PROGRAM#1 or PROGRAM#2

Refer to [Section 1.11.1](#) for automatic program selection guidelines.

Using PNS to Select PROGRAM#1 or PROGRAM#2

Using PNS to select PROGRAM#1 and inputs.

Procedure 1-14 Selecting and Starting Two Programs with PNS**Steps**

1. Set PNS inputs to a valid PROGRAM#, for example PNS0001.
2. Strobe the PNSTROB#1 input.
3. Set PNS inputs to a valid PROGRAM#2, for example PNS0002.
4. Strobe PNSTROB#2. PNS0002 is selected as PROGRAM#2.
5. Strobe PROD_START#1 to start PROGRAM#1
6. Strobe PROD_START#2 to start PROGRAM#2
7. Use HOLD#1 to pause PNS0001. Use START#1 to resume PNS0001.
8. Use HOLD#2 to pause PNS0002. Use START#2 to resume PNS0002.

Set \$MULTI_ROBO.\$HOLD_TYPE =1 to pause PNS0001 and PNS0002 separately. When you are using PNS to select your programs, keep the following in mind:

- The PNS Select method assigns PROGRAM#1 when PNSTROB#1 is used and PROGRAM#2 when PNSTROB#2 is used, unless there is a group mask conflict. Refer to [Table 1–20](#) for examples of using PNS to select programs.

Table 1–20. Using PNS to Select Programs

EXAMPLE	GROUP MASK PNS0001	GROUP MASK PNS0002	EXECUTION CASE	PNS INPUT	PNSTROBE INPUT	RESULTS OF SELECTION
1	(1,1,*,*)(Note)	(1,*,*,*)	A	PNS0001	PNSTROB	PNS0001 becomes PROGRAM#1
			B	PNS0002	PNSTROB	PNS0002 becomes PROGRAM#1
			C	PNS0001	PNSTROB2	PNS0001 becomes PROGRAM#2
			D	PNS0002	PNSTROB2	Program is not selected. "Program unsuitable" (Note 2)

Table 1–20. Using PNS to Select Programs (Cont'd)

EXAMPLE	GROUP MASK PNS0001	GROUP MASK PNS0002	EXECUTION CASE	PNS INPUT	PNSTROBE INPUT	RESULTS OF SELECTION
2	(1,1,*,*)(Note)	(1,1,*,*)	A	PNS0001	PNSTROB	PNS0001 = PROGRAM#1
			B	PNS0002	PNSTROB	PNS0002 = PROGRAM#1
			C	PNS0001	PNSTROB2	PNS0001 = PROGRAM#2
			D	PNS0002	PNSTROB2	PNS0002 = PROGRAM#2
3	(1,*,*,*)	(*,*,1,*)	A	PNS0001	PNSTROB	PNS001 = PROGRAM#1
			B	PNS0002	PNSTROB	PNS0002 = PROGRAM#1
			C	PNS0001	PNSTROB2	Program is not selected. "Program unsuitable" (Note 2)
			D	PNS0002	PNSTROB2	PNS0002 = PROGRAM#2

For each example, there is a group mask for two programs and four execution cases. You can interpret the row in the following way:

Example 1, Execution Case C=> Set PNS inputs for program PNS0001, then use PNSTROB#2 to select the program. When it is selected, PNS0001 is assigned as PROGRAM#2.

Note 1. You cannot simultaneously execute PNS0001 and PNS0002 above because the group masks of the programs will conflict. Both programs that you select for simultaneous execution must have unique group masks.

Note 2. This is a group mask conflict case.

Note By changing the jog motion group, the selected program is also changed. The changed "selected program" is displayed on program selection screen, and you can edit it in the teach pendant editor.

For example, suppose PROGRAM#1 has only group1 and PROGRAM#2 has only group2.

- By selecting jog group1, PROGRAM#1 automatically becomes selected program and the program is displayed on the edit screen.
- By selecting jog group2, PROGRAM#2 automatically becomes selected program and the program is displayed on the edit screen.
- If the "selected program" has both group1 and group2, or the "selected program" has no motion group, changing the jog motion group doesn't automatically change the "selected program". To change "selected program" in these cases, display program selection screen, set cursor to a program that you want to display and press ENTER.

If necessary, review Production Operation Setup in Chapter 6 further in this manual for detailed information about setting up PNS.

1.11.8 Selecting Programs While PROGRAM#1 and PROGRAM#2 are Running or Paused

The operation of Multi-robot control is effected by the value of \$SCR.\$MAXNUMTASK. \$SCR.\$MAXNUMTASK should be 2. The following assumes that PROGRAM#1 and PROGRAM#2 control only one robot group each (positioner axes might not be in the group mask).

When PROGRAM#1 or PROGRAM#2 is PAUSED and \$SCR.\$MAXNUMTASK = 2;

You can select another PROGRAM#1 or PROGRAM#2:

- When the teach pendant enable switch is ON, the original PROGRAM#1(#2) will be aborted, and the newly selected program will become PROGRAM#1(#2).
- When the teach pendant enable switch is OFF, you can not select another program unless you abort the current program. The message "TPIF-013 Other program is running" is displayed if you do not abort first.

When PROGRAM#1 or PROGRAM#2 is PAUSED and \$SCR.\$MAXNUMTASK >2;

- The newly selected PROGRAM#1(#2) will become PROGRAM#1(#2).
- \$MULTI_ROBO_\$TP_DEFPROG1(2) will be reassigned to the newly selected program.
- The editor will display the program that you selected but the program interpreter will still reference the original paused PROGRAM#1(#2).
- Start#1 or Start#2 will not restart the original paused program because Start#1(#2) executes \$MULTI_ROBO_\$TP_DEFPROG1(2).
- The editor does not switch between programs when the jog group is changed from group 1 to group 2.
- If you press SHIFT+FWD, the following messages will be displayed:

```
INTP-105 progrname, 1 , Run request failed  
PROG-040 Already locked by other task  
When PROGRAM#1(#2) is RUNNING and $SCR.$MAXNUMTASK = 2;
```

- Regardless of the ON/OFF state of the Teach Pendant Enable switch, when you try to select a new PROGRAM#1, the following message will be displayed:

```
TPIF-013 Other program is running  
When PROGRAM#1(#2) is RUNNING and $SCR.$MAXNUMTASK > 2;
```

- The new program will automatically become PROGRAM#1(#2).
- The editor will display the new PROGRAM#1(#2), not the currently executing program.
- The editor does not switch between programs when the jog group is changed from group 1 to group 2.
- If you press SHIFT+FWD to run the program, the following message will be displayed:

```
INTP-105 <progrname>, 1, Run request failed  
PROG-040 already locked by another task.
```

1.12 MULTI-TASKS EXECUTION BY SINGLE UOP USING MASH (MULTI ARM SHELL)

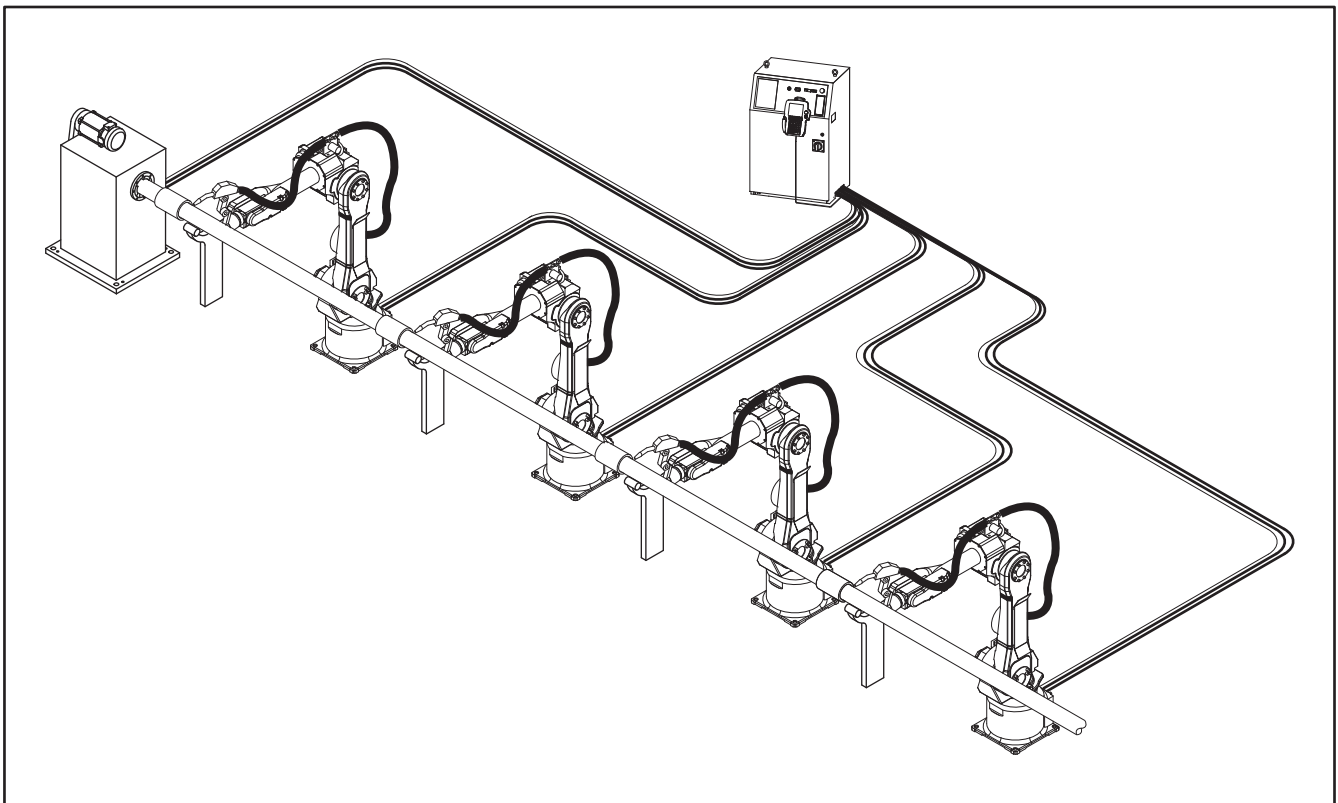
1.12.1 Overview

With multiple arms and multiple positioner groups available on a single controller, more advance processing methods can be used, for example:

- Multiple robots on a single controller could run independently, each with their own programs.
- A cell could use two robots using coordinated motion to perform part handling with one or two process robots.
- Multiple process arm(s) could be coordinated to a positioner device to perform simultaneous coordinated motion as shown in [Figure 1–6](#)
- Your processing could encompass all of the above methods.

The MultiARM system provides great flexibility to meet your processing needs.

Figure 1–6. QuadArm System with Positioner



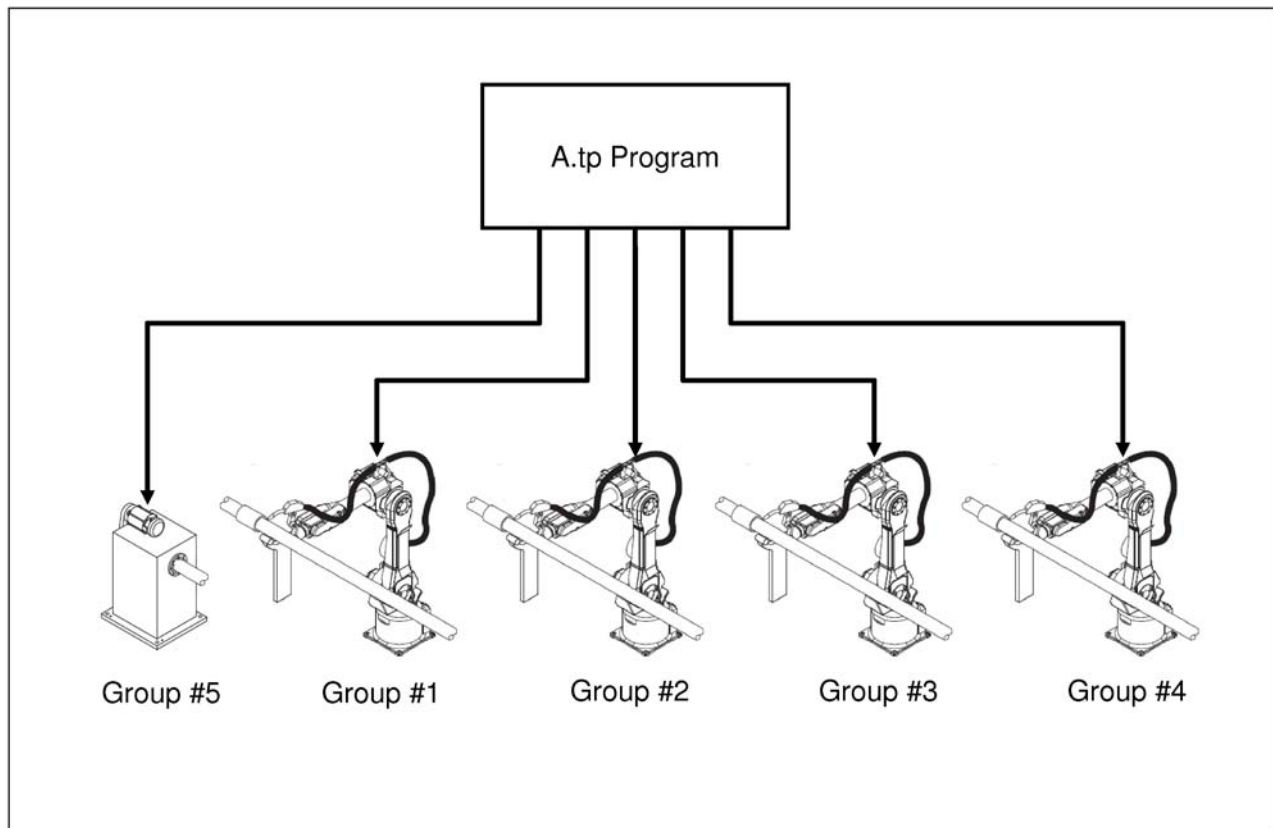
Multi-group programs can be written that include all motion groups; refer to Single Task Operation, [Section 1.12.1.1](#) . Another approach to programming is to provide multiple programs for independent robot operation; refer to Multiple Task Operation, [Section 1.12.1.2](#) . Your processing requirements will determine which programming method you should select.

You can execute these different programming methods through the standard UOP interface, the Multi-Robot Control UOP, or by using the MASH (MultiARM SHell) in combination with a UOP method.

1.12.1.1 Single Task Operation

In typical single task application, one program is used to drive all motion groups in a coordinated manner so that all arms/positioners move simultaneously (start together and end together). In addition, application processing is performed simultaneously. In this case, one program is selected and run per job.

Figure 1–7. Single Task Operation — One TP program drives all motion groups



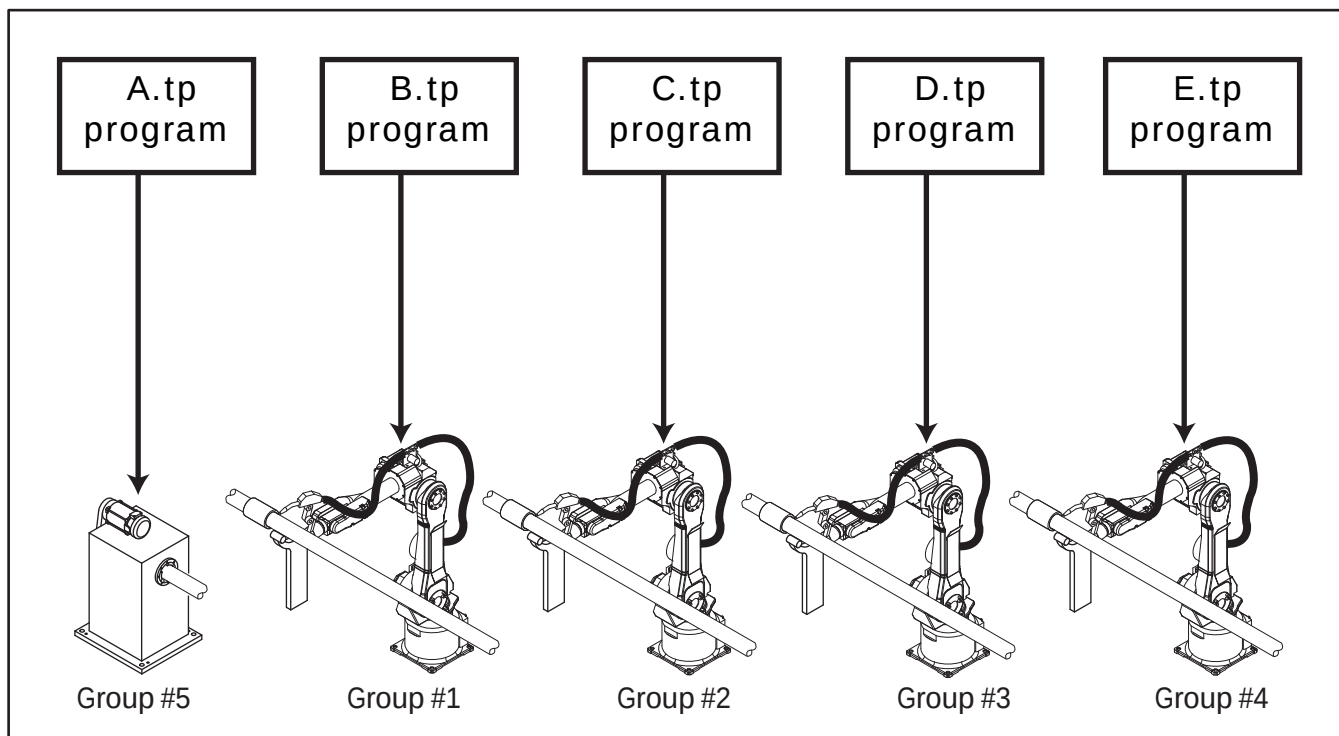
1.12.1.2 Multiple Task Operation

With the multi-task, multi-arm operation, each robot is driven by a separate task. That is, each program drives one (or more) robot arm, and multiple programs are run concurrently to carry out one job. The motions for each arm and application processing will be asynchronous (start and stops independently) in this case, as the timing of each program execution might be independent. The programs can use the “process synchronization” feature to synchronize the timing between tasks at certain points in the programs.

Multiple task execution is performed by selecting and executing **each** program that you want to have running simultaneously through a UOP method.

Note Style select does not support multiple task execution.

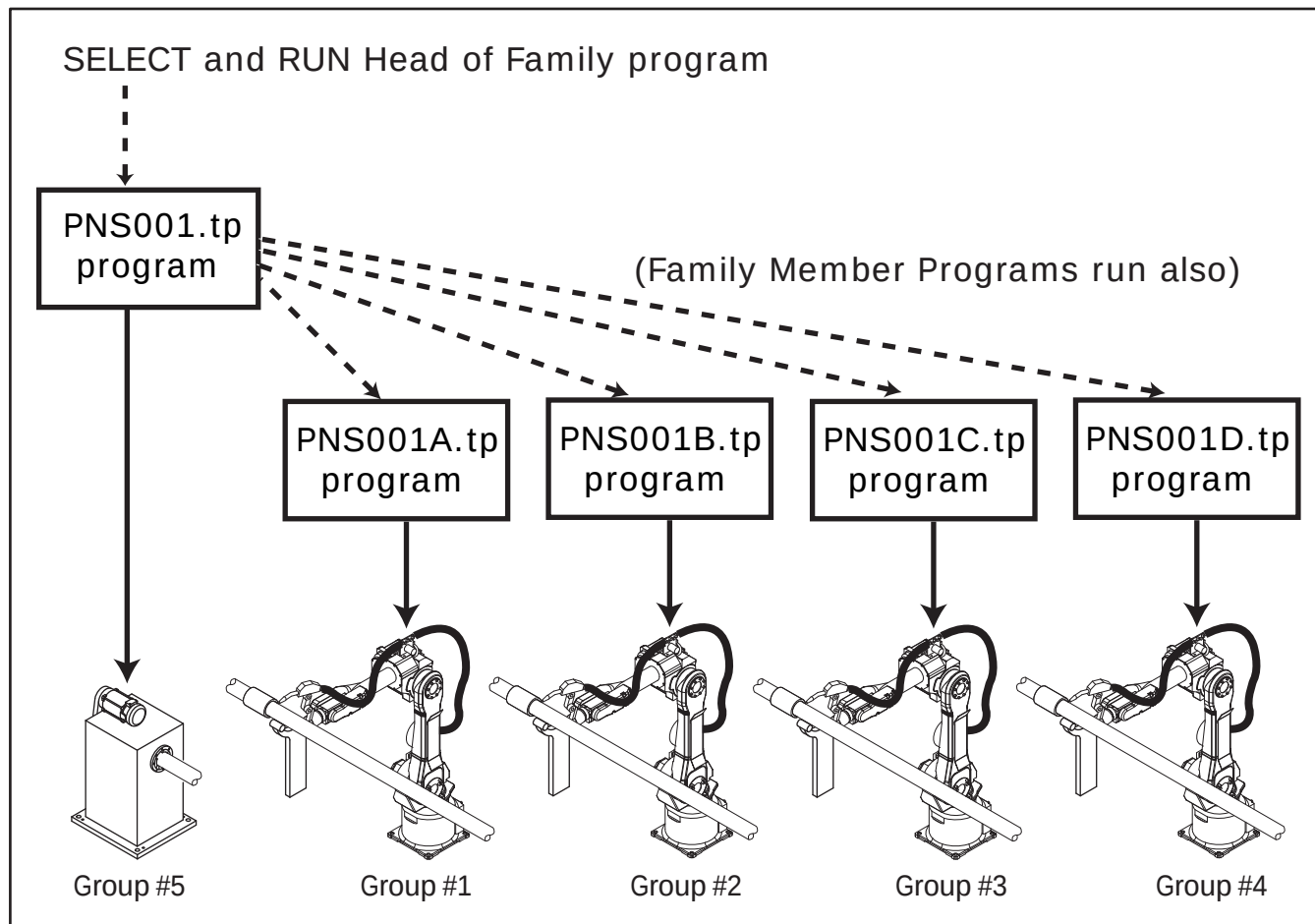
Figure 1–8. Multiple Task Operation — Concurrent execution of independent TP programs



1.12.2 MASH Feature for Multitasking Operations

To facilitate multitasking operation, the system provides a way to select and run a family of programs, so that one single interface can perform one SELECT and one RUN operation per each job.

To use the Multiple Task Operation programming style, you can define a family of programs and designate the “head of family” program and “family member” programs. If the head of family program is selected either by the teach pendant or by the PLC (via the UOP interface), all family member programs are automatically selected, and can be run all at once simply by running the head of family program, either by the teach pendant, or by the UOP interface. See [Section 1.12.4](#) for how to define a family of robot programs.

Figure 1–9. MASH Feature for Multi-Task Operations — SELECT and RUN Family of Programs

1.12.2.1 Production Operation

To use the MASH multi-tasking program feature you must first define the *head of family* robot and set up the coordination between *family members* and their respective motion groups. Then, you can create a head of family program and family member programs and set up the program header information for each program. After you have created a family of robot programs with the program headers and names properly set, the family of programs can be selected and run by selecting and running the head of the family program either from the teach pendant or from a UOP.

Style select, PNS, and RSR methods are supported for use with MultiARM shell (MASH) execution. In each case, the PLC selects and runs only the head of family program. The system will automatically select and run the family members at the time the head of family program is selected and run.

1.12.2.2 Error Recovery

For a single task operation, the resume_prog and maint_prog type error recovery features are supported as currently implemented. For the multiple task operation, a simple error recovery is supported. In other words, sending the “cycle start” signal on the UOP input will cause each robot to move to the stopped position (using the original path resume feature), and resume the program from there on.

1.12.2.3 Process Synchronization

When multiple programs are running in the multi-task environment to perform a job cooperatively, it is often required for the robot programs to wait for each other before proceeding to the next stage of operation. The “Process Synchronization” feature is provided for this purpose. Please refer to [Section 1.17](#) for how the Process Synchronization can be used.

For more information on the program instructions used with Process Synchronization, refer to the “Process Synchronization” section of the “Program Elements” chapter in your application-specific Setup and Operations manual.

1.12.2.4 Bypassing a robot

In the event of a robot within the cell is down for any reason, the BYPASS feature will allow the rest of the system to operate, thus keeping the line in production without the bypassed robot.

Note A bypassed robot can still cause error condition. It is user's responsibility to take necessary measures to suppress the error, either by clearing the cause of the error, or by setting machine-lock for the bypassed robots, etc.

When a robot is bypassed, the multi-arm shell (MASH) will isolate the bypassed robot from the system, and treat it as “non-existent”. If a family member robot is isolated, it is possible to select and run the family program using MASH. MASH will run the family of programs except for the programs of the bypassed robots. It is user's responsibility to ensure the integrity of the family of the programs. If the robot for the Head of Family program is bypassed, MASH will be unable to select or run the program. A new set of programs with a different Head of Family robot should be provided in this case.

In the case of multiple controller system, it is possible to keep the controllers of the bypassed robots turned off.

A robot can be bypassed in the SETUP/MASH screen. Changing any item in this menu will require the power of all controllers to be cycled. Use [Procedure 1-15](#) to bypass a robot.

Note A program of a bypassed robot could still be selected and run by the teach pendant or by the UOP input. The above discussion only applies to selecting/running a program under MASH control.

1.12.3 Setting Up MASH

1.12.3.1 MASH Setup

The MASH SETUP screen sets up and defines the multi-robot system, and establishes the robot number definition in the system. In this screen you need to specify the hostname of the controller where the robot resides and the group number where the robot is connected. The HostName is set up in the HostComm setup screen in the TCP/IP setup. If you are not using TCP/IP feature, a default name will be provided.

Use [Procedure 1-15](#) to set up MASH.

Table 1–21. MASH Setup Items

ITEM	DESCRIPTION
Bypassed Robots Output	Each bit of the group output indicates a bypassed robot, the least significant bit (LSB) being the robot #1.
Robot Number Definition	These items define the robot number within the multi-arm configuration, by specifying the following. • Hostname of the controller (in the case of multiple controller). Leave it blank in the case of a single controller configuration. • Group Number for this robot • Program name suffix to be used for this robot • Whether this robot is bypassed or not. Changing any item in the setup requires a COLD start (re-power).

Procedure 1-15 Setting Up MASH

1. Press MENU.
2. Select SETUP.
3. Select MASH. You will see a screen similar to the following.

```

MULTI-ARM SETUP
Multi-Arm Configuration                2/17
  1 Bypassed Robots Output  GO[  0]
    Robot Hostname      Grp# Suffix Bypass
  2   #1                1    Z    NO
  3   #2                2    A    NO
  4   #3                3    B    NO
  5   #4                0    C    NO
  6   #5                0    D    NO
  7   #6                0    E    NO
  8   #7                0    F    NO
  9   #8                0    G    NO

```

4. Move the cursor to Bypassed Robots Output, and set the output number.
5. Define the robot combinations and group numbers. Refer to [Table 1–21](#).

Note This shows the default name as Robot#1. Robot #2 and Robot #3 implicitly are assigned when items #2 and #3 have the group specified. You do not have to assign all groups on a controller, just those that you will be using in head of family or family member programming.

Note If setting up a multiple-controller system, be sure to setup the network configuration, including RIPE, before setting up multi-arm (MASH) configuration. (A re-power may be required). Once the network is configured, setting up MASH on a controller will set up MASH on all controllers, except for “Bypassed Robots Output” setting, which is local to a controller.

Example 1: Quad-arm system with a positioner

```

MULTI-ARM SETUP
Multi-Arm Configuration                2/17
  1 Bypassed Robots Output  GO[  0]
    Robot Hostname      Grp# Suffix Bypass
  2   #1                1    A    NO
  3   #2                2    B    NO
  4   #3                3    C    NO
  5   #4                4    D    NO
  6   #5                5    Z    NO
  7   #6                0    E    NO
  8   #7                0    F    NO
  9   #8                0    G    NO

```

In a multi-arm system with a positioner, it is recommended a program for the positioner to be the Head of Family program. The Family Member programs will control the robot arms. Accordingly the suffix for the group #1 through #4 are set A through D to represent the family members. See [Figure 1–9](#) for an example of family of programs.

Example 2: Quad-arm system with No Positioner

```

MULTI-ARM SETUP
Multi-Arm Configuration                2/17
1 Bypassed Robots Output  GO[ 0]
  Robot Hostname      Grp#  Suffix Bypass
2   #1                1    A    NO
3   #2                2    B    NO
4   #3                3    C    NO
5   #4                4    D    NO
6   #5               -1    Z    NO
7   #6                0    E    NO
8   #7                0    F    NO
9   #8                0    G    NO

```

Please note the special value of group number “-1” for the robot #5, which is a “logical” robot with NO arm associated. You may assign a Head of Family to a program with no motion group. To represent a “robot” for such no-motion program, a special value “-1” is used as the motion group number. The advantages of using a non-motion “Head of Family” program are,

- All programs to control robot arms are defined as “Family Member”, and are handled equally.
- It is possible during production operation to switch between simultaneous/coordinated motion (controlled by one single program) and independent motion (controlled by multiple programs)
- During program development, it is possible to test-run a “Family Member” program individually. It is done by selecting a family member program and run it by itself.

1.12.4 Defining a Family of Robots

A robot must be designated as part of a family by naming it according to a specified naming convention. “Job suffix” (string[1]) is set up on each defined robot, which is appended to the program name, so that the program names can be differentiated within a controller.

For example, Job suffix is defined as follows. ”

- Robot #1: “Z”
- Robot #2: “A”
- Robot #3: “B”
- Robot #4: “C”

If a program called “PQRSTUV” is selected for the robot #1 as the head of family, the following will be the program name for each of the family members:

- Robot #1: “PQRSTUV” (the head of family program name is always unchanged)

- Robot #2: “PQRSTUVA”
- Robot #3: “PQRSTUVB”
- Robot #4: “PQRSTUVC”

MultiARM Program Header Settings

The Program Header for each program will indicate the status of a program: whether it is a Head of the Family or a Family member. For the head of the family program, the header contains the information about its family members. For a family member program, the header contains the information to verify its head of the family robot. Refer to chapter *Program Elements*, section “Program Header Information”, subsections “Program Type” through “Family Members” in your application-specific *Setup and Operations Manual* for more information. See the following screen for a program header example for a head of family program.

```
Multi-Arm program data          1/3

1 Program type:    Main Program
2 Multi-Arm Main? Yes, Head of Famil
3 Family Members: [1,1,_,_,_,_,_,_]
```

Note The Family Members mask refers to the information in [Procedure 1-15](#). Each bit corresponds to a robot from the Robot number definition list.

See the following screen for a program header example for a family member program.

```
Multi-Arm program data          1/3

1 Program type:    Main Program
2 Multi-Arm Main? Yes, Family member
3 Head of Family:  Robot#1
```

After you have created a family of robot programs with the program headers and names properly set, the family of programs can be selected and run by selecting and running the “head of family” program from the teach pendant or UOP.

Note The head of family does not have to be robot #1.

1.13 TEACHING AND MODIFYING POSITIONS

Teaching program positions is done by selecting a program, entering the editor, and selecting standard motion instructions. For the Multi-Robot case, the group mask shown on the teach pendant status

line must be included in the group mask of the program otherwise an error will be posted when you attempt to record the position. "TPIF-044 Program is unsuitable for robot" will be posted in this case.

Use [Procedure 1-16](#) to teach and modify positions.

Procedure 1-16 Teaching and Modifying Positions

Steps

1. To modify position data, select the proper jog group.
2. Move the robot or other motion group to the correct position.
3. Move the cursor to the corresponding position in your program and press and hold SHIFT.
4. To record the current position as new position data, press F5, TOUCHUP. Alternatively, you can use the F1 through F4 keys to add motion instructions to your program.

All motion groups in the program will have position data updated unless a robot group in the program is currently in isolate mode.

Note When you are editing a multi-group program containing multiple robot arms, if the robot-isolation key for Robot 1 (group1) or for Robot 2 (group2) or Robot 3 or Robot 4 is set to isolate, the position data corresponding to the "isolated" group will not be changed (or recorded) when touching up an existing position (or recording a new position.)

Note If none of the Robot Isolate switches are DISABLED, the data will be touched up for all groups controlled by the switches in the current program. Make sure that all affected motion groups are in the correct location before you touchup or train a new point.

5. After a position is taught,
 - When a multi-robot program (Type 2 or Type 4) position is edited, and robot isolation is ON for one of the robots, one of the following messages will be displayed:
 - "Position has been recorded to P[1](G:1)," **when robot 2 is isolated.**
 - "Position has been recorded to P[1](G:2)," **when robot 1 is isolated.**

Note Non-robot groups assigned to Robot Isolate keyswitches will not be noted in the "recorded" alarm message.

Note Refer to [Section 1.7.2](#) for additional information regarding the Robot Isolation feature.

1.14 STOPPING PROGRAM EXECUTION

There are several ways to stop Multi Robot programs from running. Programs can be stopped by:

- SOP E-STOP, external E-STOP, or Fence input
- Loss of UOP signals ENABLE, CSTOPI, SFSPD, HOLD#1(#2)
- Teach pendant HOLD, releasing the SHIFT key during teach pendant execution, ABORT ALL in the FCTN menu
- Pause or abort severity errors in program execution
- Cycling power OFF

1.14.1 Pausing a Program

PAUSE suspends motion and welding operation but allows for restarting the program and weld process. When a program is paused, you can jog the robot from the paused position (to repair the torch, for example), and via "Return to Path feature" return the robot to the fault position automatically and then restart motion and welding. Paused programs retain control of motion groups, PROGRAM#1(2) assignment (when \$SCR.\$MAXNUMTASK =2), and weld equipment.

1.14.2 Pause Program from the Teach Pendant

When the HOLD key on the teach pendant is pressed, all running programs are paused.

Refer to Section "Selecting Programs While PROGRAM#1 and PROGRAM#2 are Running or Paused" for information about restarting paused programs after a new program has been selected.

1.14.3 Pause Program from the UOP

When HOLD or HOLD2 is turned off,

- When \$MULTI_ROBO.\$HOLD_TYPE = 0, if either HOLD or HOLD2 is turned OFF, all executing programs are paused.
- When \$MULTI_ROBO.\$HOLD_TYPE = 1, if HOLD is OFF, "PROGRAM#1" is paused. If HOLD2 is OFF, "PROGRAM#2" is paused. To pause additional running (multi-tasking) programs, turn the ENBL input OFF.
- Loss of UOP signals ENABLE, CSTOPI, and SFSPD cause all programs to PAUSE.

Refer to Section "Selecting Programs While PROGRAM#1 and PROGRAM#2 are Running or Paused" for information about restarting paused programs after a new program has been selected.

Note HOLD and HOLD2 signals are normally high.

1.15 ABORTING PROGRAMS

Aborting programs stops the robot motion and application process. When a program is aborted, it cannot recover the weld process when restarted. The program releases control of motion groups it was using and also releases control of application equipment. If you restart the program, the robot will not return to the aborted position if jogged away (motion will be planned from the current position to the next trained position.). The weld will not start until the next process “Start” instruction is executed. The aborted program can be immediately de-assigned as PROGRAM#1(2).

1.15.1 Aborting Programs from the Teach Pendant

You can abort all running programs from the teach pendant using [Procedure 1-17](#) .

Procedure 1-17 Procedure 21 Aborting Programs from the Teach Pendant

Steps

1. To abort all programs from the teach pendant, press FCTN.
2. Select Abort All.
3. Press ENTER.

1.15.2 Aborting Programs from the UOP

If ‘CSTOPI for ABORT’ in System Configuration menu is TRUE, programs are aborted when CSTOPI turns on. The default setting of ‘CSTOPI for ABORT’ is FALSE, ‘Abort all programs by CSTOPI’ setting defines the function of CSTOPI in the following way:

- When **Abort all programs by CSTOPI** is **FALSE** (Default), only the selected program is aborted by **CSTOPI** .
- When **Abort all programs from CSTOPI** is **TRUE** , all programs are aborted by **CSTOPI**
- **CSTOPI** is configured in the menus/system/config menu, “CSTOPI for ABORT:” item.

Note The recommended setting for **Abort all programs from CSTOPI** is TRUE for Multi-Robot.

1.15.3 Aborting Programs by Cycling Controller Power

If the power is removed from the controller while programs are executing:

- When \$SEMIPOWERFL = FALSE, all executing programs are aborted
- When \$SEMIPOWERFL = TRUE, all executing programs are in paused state when the controller is turned on. Program Type 1 through Type 4 will recover. Program Type 5 (executing master and slave programs) will not recover.

1.16 COORDINATED MOTION SUPPORT

1.16.1 Standard Coordinated Motion

Standard coordinated motion is used in Type 3 programs (two robots and a positioner in the same program) when you perform coordinated motion between a leader group (positioner) and two follower groups (robots). In Multi-Arm systems, two robots may be required to produce one part or to produce two similar or asymmetric parts in the same positioner tooling. Paths on these parts are the same length. Both robots execute the same or mirrored motion while the positioner moves the parts. The relative TCP speed of one robot will be the programmed speed; the other robot TCP will be close to the programmed speed if the path segments are close to being the same length.

Refer to Standard coordinated motion is used in Type 3 programs (two robots between a leader group (positioner) and two follower groups (robots). In MultiARM speed if the path segments are close to being the same length. A separate program might be required to control weld equipment #2, using the multi-tasking feature.

Refer to [Section 1.2.7](#) , “MultiARM Programming Methods” for more information about program types.

1.16.2 Robot to Robot Coordinated Motion

Note Refer to the *Coordinated Motion Setup and Operations Manual* for more information.

Robot to robot coordinated motion is used for material handling is the method for CD pairs with a known kinematics leader group such as a to the leader robot origin.

FANUC recommends that high accuracy calibration between robots and positions be performed for material handling situations, especially multiple arms transferring a single part by using iRCalibration Vision MultiCAL option, to accurately define the robot to robot calibration

Refer to the *Coordinated Motion Setup and Operations Manual* for more information.

1.17 PROCESS SYNCHRONIZATION

1.17.1 Overview

Process Synchronization is a feature for multiple programs to synchronize with each other within the programs. For more information, refer to the application-specific *Setup and Operations Manual*.

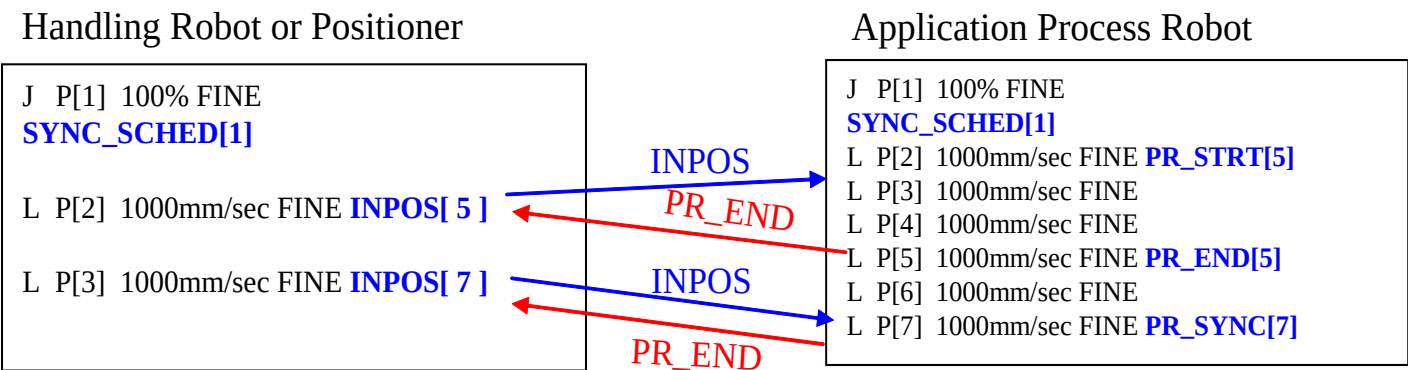
1.17.2 Features

Process Synchronization utilizes the INPOS / PR_START / PR_END /PR_SYNC instructions within the program. The feature is similar to what “semaphore” and “wait semaphore” would provide, and is made easier for the user to program. Also the process synchronization instruction is attached to a motion statement so that the synchronization takes place at end of the motion interval.

1.17.3 Program Example

See the following programming example.

Figure 1–10. Process Synchronization Example



In this example, the process robot waits for the handling robot to come in position (at PR_STRT[5]). When the handling robot comes in position, INPOS instruction will “signal” the process robot, to cause the process robot to proceed to the next instruction. At the same time, the handling robot will wait for the process robot to finish. When the process robot comes to PR_END instruction, this will “signal” the handling robot, to cause both robots to proceed to the next instruction.

PR_SYNC instruction functions as combination of PR_STRT and PR_END instructions in one line.

1.17.3.1 Program Elements

Process Synchronization uses the following program instructions.

- SYNC_SCHED[sch_n]
- INPOS[seq_n]
- PR_STRT[seq_n]
- PR_END[seq_n]
- PR_SYNC[seq_n]

For more information on the program instructions, refer to the “Process Synchronization” section of the “Program Elements” chapter in your application-specific Setup and Operations manual.

1.17.4 Limitations

When used with multiple process robots, PR_STRT and PR_SYNC instructions may not be used at the same time. That is, all process robots must use PR_STRT/PR_END instructions, or all process robot use PR_SYNC instructions. It is not allowed to have some process robot use PR_STRT/PR_END instructions while other process robots use PR_SYNC instruction

MULTI ARM TOUCH SENSING (OPTIONAL FEATURE)

Contents

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2.2	System Setup	2-2
2.3	Limitations	2-2

2.1 Overview

The existing touch sensing system supports Multi-Arm systems with the following limitations:

- In order to support asymmetrical touch sensing concurrently, two TP programs are required:
 - one TP program with 2D search for the first robot
 - another TP program with 3D search for the second robot

For two coordinated TP programs to share the same leader, sequential search execution is mandatory due to PG motion group control. TP program conversion is required prior to touch offset application.

- Only one touch sensing process robot is allowed per TP program.

The new simultaneous touch sensing system overcomes prior shortcomings with the following enhancements to support the multi-arm controller:

- In symmetrical touch sensing, which accounts for about 90% of all multi-arm touch sensing applications, simultaneous search motions for all process robots are available with one search instruction in a single TP program.
- In symmetrical touch sensing, touch offsets for all process robots are saved in a single position register PR[.]. There is no need to convert TP programs via transform utility.

Note For information on Touch Sensing for individual (not multi-arm) robots, refer to your application-specific *Setup and Operations manual*.

Note In Head of family/Family member programming, the touch sensing is performed in each family program individually.

2.2 System Setup

Multi-Arm Simultaneous Touch Sensing can be configured via two simple steps:

1. Include all required touch process robots in the robot group mask item of the specified touch schedule.
2. Configure the touch frame for each included touch process robot.

2.3 Limitations

Simultaneous touch sensing in a single TP program is available for the multi-arm controller with the following limitations, including:

- Multiple process robots share the same touch schedule with the same search pattern and pattern type.
- Used for symmetrical touch sensing only.

For asymmetrical touch sensing, you must use multiple TP programs with different search patterns, and conversion of touch offset TP programs is required for two coordinated programs sharing the same leader.

Glossary

A

abort

Abnormal termination of a computer program caused by hardware or software malfunction or operator cancellation.

absolute pulse code system

A positional information system for servomotors that relies on battery-backed RAM to store encoder pulse counts when the robot is turned off. This system is calibrated when it is turned on.

A/D value

An analog to digital-value. Converts a multilevel analog electrical system pattern into a digital bit.

AI

Analog input.

AO

Analog output.

alarm

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

algorithm

A fixed step-by-step procedure for accomplishing a given result.

alphanumeric

Data that are both alphabetical and numeric.

AMPS

Amperage amount.

analog

The representation of numerical quantities by measurable quantities such as length, voltage or resistance. Also refers to analog type I/O blocks and distinguishes them from discrete I/O blocks. Numerical data that can vary continuously, for example, voltage levels that can vary within the range of -10 to +10 volts.

AND

An operation that places two contacts or groups of contacts in series. All contacts in series control the resulting status and also mathematical operator.

ANSI

American National Standard Institute, the U.S. government organization with responsibility for the development and announcement of technical data standards.

APC

See absolute pulse code system.

APC motor

See servomotor.

application program

The set of instructions that defines the specific intended tasks of robots and robot systems to make them reprogrammable and multifunctional. You can initiate and change these programs.

arm

A robot component consisting of an interconnecting set of links and powered joints that move and support the wrist socket and end effector.

articulated arm

A robot arm constructed to simulate the human arm, consisting of a series of rotary motions and joints, each powered by a motor.

ASCII

Abbreviation for American Standard Code for Information Interchange. An 8-level code (7 bits plus 1 parity bit) commonly used for the exchange of data.

automatic mode

The robot state in which automatic operation can be initiated.

automatic operation

The time during which robots are performing programmed tasks through unattended program execution.

axis

1. A straight line about which a robot joint rotates or moves. 2. One of the reference lines or a coordinate system. 3. A single joint on the robot arm.

B**backplane**

A group of connectors mounted at the back of a controller rack to which printed circuit boards are mated.

BAR

A unit of pressure equal to 100,000 pascals.

barrier

A means of physically separating persons from the restricted work envelope; any physical boundary to a hazard or electrical device/component.

battery low alarm

A programmable value (in engineering units) against which the analog input signal automatically is compared on Genius I/O blocks. A fault is indicated if the input value is equal to or less than the low alarm value.

baud

A unit of transmission speed equal to the number of code elements (bits) per second.

big-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

binary

A numbering system that uses only 0 and 1.

bit

Contraction of binary digit. 1. The smallest unit of information in the binary numbering system, represented by a 0 or 1. 2. The smallest division of a programmable controller word.

bps

Bits per second.

buffer

A storage area in the computer where data is held temporarily until the computer can process it.

bus

A channel along which data can be sent.

bus controller

A Genius bus interface board for a programmable controller.

bus scan

One complete communications cycle on the serial bus.

Bus Switching Module

A device that switches a block cluster to one bus or the other of a dual bus.

byte

A sequence of binary digits that can be used to store a value from 0 to 255 and usually operated upon as a unit. Consists of eight bits used to store two numeric or one alpha character.

C**calibration**

The process whereby the joint angle of each axis is calculated from a known reference point.

Cartesian coordinate system

A coordinate system whose axes (x, y, and z) are three intersecting perpendicular straight lines. The origin is the intersection of the axes.

Cartesian coordinates

A set of three numbers that defines the location of a point within a rectilinear coordinate system and consisting of three perpendicular axes (x, y, z).

cathode ray tube

A device, like a television set, for displaying information.

central processing unit

The main computer component that is made up of a control section and an arithmetic-logic section. The other basic units of a computer system are input/output units and primary storage.

channel

The device along which data flow between the input/output units of a computer and primary storage.

character

One of a set of elements that can be arranged in ordered groups to express information. Each character has two forms: 1. a man-intelligible form, the graphic, including the decimal digits 0-9, the letters A-Z, punctuation marks, and other formatting and control symbols; 2. a computer intelligible form, the code, consisting of a group of binary digits (bits).

circular

A MOTYPE option in which the robot tool center point moves in an arc defined by three points. These points can be positions or path nodes.

clear

To replace information in a storage unit by zero (or blank, in some machines).

closed loop

A control system that uses feedback. An open loop control system does not use feedback.

C-MOS RAM

Complementary metal-oxide semiconductor random-access memory. A read/write memory in which the basic memory cell is a pair of MOS (metal-oxide semiconductor) transistors. It is an implementation of S-RAM that has very low power consumption, but might be less dense than other S-RAM implementations.

coaxial cable

A transmission line in which one conductor is centered inside and insulated from an outer metal tube that serves as the second conductor. Also known as coax, coaxial line, coaxial transmission line, concentric cable, concentric line, concentric transmission line.

component

An inclusive term used to identify a raw material, ingredient, part or subassembly that goes into a higher level of assembly, compound or other item.

computer

A device capable of accepting information, applying prescribed processes to the information, and supplying the results of these processes.

configuration

The joint positions of a robot and turn number of wrist that describe the robot at a specified position. Configuration is designated by a STRING value and is included in positional data.

continuous path

A trajectory control system that enables the robot arm to move at a constant tip velocity through a series of predefined locations. A rounding effect of the path is required as the tip tries to pass through these locations.

continuous process control

The use of transducers (sensors) to monitor a process and make automatic changes in operations through the design of appropriate feedback control loops. While such devices historically have been mechanical or electromechanical, microcomputers and centralized control is now used, as well.

continuous production

A production system in which the productive equipment is organized and sequenced according to the steps involved to produce the product. Denotes that material flow is continuous during the production process. The routing of the jobs is fixed and set-ups are seldom changed.

controlled stop

A controlled stop controls robot deceleration until it stops. When a safety stop input such as a safety fence signal is opened, the robot decelerates in a controlled manner and then stops. After the robot stops, the Motor Control Contactor opens and drive power is removed.

controller

A hardware unit that contains the power supply, operator controls, control circuitry, and memory that directs the operation and motion of the robot and communications with external devices. See control unit.

controller memory

A medium in which data are retained. Primary storage refers to the internal area where the data and program instructions are stored for active use, as opposed to auxiliary or external storage (magnetic tape, disk, diskette, and so forth.)

control, open-loop

An operation where the computer applies control directly to the process without manual intervention.

control unit

The portion of a computer that directs the automatic operation of the computer, interprets computer instructions, and initiates the proper signals to the other computer circuits to execute instructions.

coordinate system

See Cartesian coordinate system.

CPU

See central processing unit.

CRT

See cathode ray tube.

cps (viscosity)

Centipoises per second.

CRT/KB

Cathode ray tube/keyboard. An optional interface device for the robot system. The CRT/KB is used for some robot operations and for entering programs. It can be a remote device that attaches to the robot via a cable.

cycle

1. A sequence of operations that is repeated regularly. The time it takes for one such sequence to occur. 2. The interval of time during which a system or process, such as seasonal demand or a manufacturing operation, periodically returns to similar initial conditions. 3. The interval of time during which an event or set of events is completed. In production control, a cycle is the length of time between the release of a manufacturing order and shipment to the customer or inventory.

cycle time

1. In industrial engineering, the time between completion of two discrete units of production. 2. In materials management, the length of time from when material enters a production facility until it exits. See throughput.

cursor

An indicator on a teach pendant or CRT display screen at which command entry or editing occurs. The indicator can be a highlighted field or an arrow (> or ^).

cylindrical

Type of work envelope that has two linear major axes and one rotational major axis. Robotic device that has a predominantly cylindrical work envelope due to its design. Typically has fewer than 6 joints and typically has only 1 linear axis.

D**D/A converter**

A digital-to-analog converter. A device that transforms digital data into analog data.

D/A value

A digital-to-analog value. Converts a digital bit pattern into a multilevel analog electrical system.

daisy chain

A means of connecting devices (readers, printers, etc.) to a central processor by party-line input/output buses that join these devices by male and female connectors. The last female connector is shorted by a suitable line termination.

daisy chain configuration

A communications link formed by daisy chain connection of twisted pair wire.

data

A collection of facts, numeric and alphabetical characters, or any representation of information that is suitable for communication and processing.

data base

A data file philosophy designed to establish the independence of computer program from data files. Redundancy is minimized and data elements can be added to, or deleted from, the file designs without changing the existing computer programs.

DC

Abbreviation for direct current.

DEADMAN switch

A control switch on the teach pendant that is used to enable servo power. Pressing the DEADMAN switch while the teach pendant is on activates servo power and releases the robot brakes; releasing the switch deactivates servo power and applies the robot brakes.

debugging

The process of detecting, locating and removing mistakes from a computer program, or manufacturing control system. See diagnostic routine.

deceleration tolerance

The specification of the percentage of deceleration that must be completed before a motion is considered finished and another motion can begin.

default

The value, display, function or program automatically selected if you have not specified a choice.

deviation

Usually, the absolute difference between a number and the mean of a set of numbers, or between a forecast value and the actual data.

device

Any type of control hardware, such as an emergency-stop button, selector switch, control pendant, relay, solenoid valve, or sensor.

diagnostic routine

A test program used to detect and identify hardware/software malfunctions in the controller and its associated I/O equipment. See debugging.

diagnostics

Information that permits the identification and evaluation of robot and peripheral device conditions.

digital

A description of any data that is expressed in numerical format. Also, having the states On and Off only.

digital control

The use of a digital computer to perform processing and control tasks in a manner that is more accurate and less expensive than an analog control system.

digital signal

A single point control signal sent to or from the controller. The signal represents one of two states: ON (TRUE, 1. or OFF (FALSE, 0).

directory

A listing of the files stored on a device.

discrete

Consisting of individual, distinct entities such as bits, characters, circuits, or circuit components. Also refers to ON/OFF type I/O blocks.

disk

A secondary memory device in which information is stored on a magnetically sensitive, rotating disk.

disk memory

A non-programmable, bulk-storage, random-access memory consisting of a magnetized coating on one or both sides of a rotating thin circular plate.

drive power

The energy source or sources for the robot servomotors that produce motion.

DRAM

Dynamic Random Access Memory. A read/write memory in which the basic memory cell is a capacitor. DRAM (or D-RAM) tends to have a higher density than SRAM (or S-RAM). Due to the support circuitry required, and power consumption needs, it is generally impractical to use. A battery can be used to retain the content upon loss of power.

E**edit**

1. A software mode that allows creation or alteration of a program. 2. To modify the form or format of data, for example, to insert or delete characters.

emergency stop

The operation of a circuit using hardware-based components that overrides all other robot controls, removes drive power from the actuators, and causes all moving parts of to stop. The operator panel and teach pendant are each equipped with EMERGENCY STOP buttons.

enabling device

A manually operated device that, when continuously activated, permits motion. Releasing the device stops the motion of the robot and associated equipment that might present a hazard.

encoder

1. A device within the robot that sends the controller information about where the robot is. 2. A transducer used to convert position data into electrical signals. The robot system uses an incremental optical encoder to provide position feedback for each joint. Velocity data is computed from the encoder signals and used as an additional feedback signal to assure servo stability.

end effector

An accessory device or tool specifically designed for attachment to the robot wrist or tool mounting plate to enable the robot to perform its intended tasks. Examples include gripper, spot weld gun, arc weld gun, spray paint gun, etc.

end-of-arm tooling

Any of a number of tools, such as welding guns, torches, bells, paint spraying devices, attached to the faceplate of the robot wrist. Also called end effector or EOAT.

engineering units

Units of measure as applied to a process variable, for example, psi, Degrees F., etc.

envelope, maximum

The volume of space encompassing the maximum designed movements of all robot parts including the end effector, workpiece, and attachments.

EOAT

See end of arm tooling, tool.

EPROM

Erasable Programmable Read Only Memory. Semiconductor memory that can be erased and reprogrammed. A non-volatile storage memory.

error

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

error message

A numbered message, displayed on the CRT/KB and teach pendant, that indicates a system problem or warns of a potential problem.

Ethernet

A Local Area Network (LAN) bus-oriented, hardware technology that is used to connect computers, printers, terminal concentrators (servers), and many other devices together. It consists of a master cable and connection devices at each machine on the cable that allow the various devices to "talk" to each other. Software that can access the Ethernet and cooperate with machines connected to the cable is necessary. Ethernets come in varieties such as baseband and broadband and can run on different media, such as coax, twisted pair and fiber. Ethernet is a trademark of Xerox Corporation.

execute

To perform a specific operation, such as one that would be accomplished through processing one statement or command, a series of statements or commands, or a complete program or command procedure.

extended axis

An optional, servo-controlled axis that provides extended reach capability for a robot, including in-booth rail, single- or double-link arm, also used to control motion of positioning devices.

F**faceplate**

The tool mounting plate of the robot.

feedback

1. The signal or data fed back to a commanding unit from a controlled machine or process to denote its response to the command signal. The signal representing the difference between actual response and desired response that is used by the commanding unit to improve performance of the controlled machine or process. 2. The flow of information back into the control system so that actual performance can be compared with planned performance, for instance in a servo system.

field

A specified area of a record used for a particular category of data. 2. A group of related items that occupy the same space on a CRT/KB screen or teach pendant LCD screen. Field name is the name of the field; field items are the members of the group.

field devices

User-supplied devices that provide information to the PLC (inputs: push buttons, limit switches, relay contacts, and so forth) or perform PLC tasks (outputs: motor starters, solenoids, indicator lights, and so forth.)

file

1. An organized collection of records that can be stored or retrieved by name. 2. The storage device on which these records are kept, such as bubble memory or disk.

filter

A device to suppress interference that would appear as noise.

Flash File Storage

A portion of FROM memory that functions as a separate storage device. Any file can be stored on the FROM disk.

Flash ROM

Flash Read Only Memory. Flash ROM is not battery-backed memory but it is non-volatile. All data in Flash ROM is saved even after you turn off and turn on the robot.

flow chart

A systems analysis tool to graphically show a procedure in which symbols are used to represent operations, data, flow, and equipment. See block diagram, process chart.

flow control

A specific production control system that is based primarily on setting production rates and feeding work into production to meet the planned rates, then following it through production to make sure that it is moving. This concept is most successful in repetitive production.

format

To set up or prepare a memory card or floppy disk (not supported with version 7.20 and later) so it can be used to store data in a specific system.

FR

See Flash ROM.

F-ROM

See Flash ROM.

FROM disk

See Flash ROM.

G

general override stat

A percentage value that governs the maximum robot jog speed and program run speed.

Genius I/O bus

The serial bus that provides communications between blocks, controllers, and other devices in the system especially with respect to GE FANUC Genius I/O.

gripper

The "hand" of a robot that picks up, holds and releases the part or object being handled. Sometimes referred to as a manipulator. See EOAT, tool.

group signal

An input/output signal that has a variable number of digital signals, recognized and taken as a group.

gun

See applicator.

H

Hand Model.

Used in Interference Checking, the Hand Model is the set of virtual model elements (spheres and cylinders) that are used to represent the location and shape of the end of arm tooling with respect to the robot's faceplate.

hardware

1. In data processing, the mechanical, magnetic, electrical and electronic devices of which a computer, controller, robot, or panel is built. 2. In manufacturing, relatively standard items such as nuts, bolts, washers, clips, and so forth.

hard-wire

To connect electric components with solid metallic wires.

hard-wired

1. Having a fixed wired program or control system built in by the manufacturer and not subject to change by programming. 2. Interconnection of electrical and electronic devices directly through physical wiring.

hazardous motion

Unintended or unexpected robot motion that can cause injury.

hexadecimal

A numbering system having 16 as the base and represented by the digits 0 through 9, and A through F.

hold

A smoothly decelerated stopping of all robot movement and a pause of program execution. Power is maintained on the robot and program execution generally can be resumed from a hold.

HTML.

Hypertext Markup Language. A markup language that is used to create hypertext and hypermedia documents incorporating text, graphics, sound, video, and hyperlinks.

http.

Hypertext transfer protocol. The protocol used to transfer HTML files between web servers.

I**impedance**

A measure of the total opposition to current flow in an electrical circuit.

incremental encoder system

A positional information system for servomotors that requires calibrating the robot by moving it to a known reference position (indicated by limit switches) each time the robot is turned on or calibration is lost due to an error condition.

index

An integer used to specify the location of information within a table or program.

index register

A memory device containing an index.

industrial robot

A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions in order to perform a variety of tasks.

industrial robot system

A system that includes industrial robots, end effectors, any equipment devices and sensors required for the robot to perform its tasks, as well as communication interfaces for interlocking, sequencing, or monitoring the robot.

information

The meaning derived from data that have been arranged and displayed in a way that they relate to that which is already known. See data.

initialize

1. Setting all variable areas of a computer program or routine to their desired initial status, generally done the first time the code is executed during each run. 2. A program or hardware circuit that returns a program a system, or hardware device to an original state. See startup, initial.

input

The data supplied from an external device to a computer for processing. The device used to accomplish this transfer of data.

input device

A device such as a terminal keyboard that, through mechanical or electrical action, converts data from the form in which it has been received into electronic signals that can be interpreted by the CPU or programmable controller. Examples are limit switches, push buttons, pressure switches, digital encoders, and analog devices.

input processing time

The time required for input data to reach the microprocessor.

input/output

Information or signals transferred between devices, discreet electrical signals for external control.

input/output control

A technique for controlling capacity where the actual output from a work center is compared with the planned output developed by CRP. The input is also monitored to see if it corresponds with plans so that work centers will not be expected to generate output when jobs are not available to work on.

integrated circuit

A solid-state micro-circuit contained entirely within a chip of semiconductor material, generally silicon. Also called chip.

interactive

Refers to applications where you communicate with a computer program via a terminal by entering data and receiving responses from the computer.

interface

1. A concept that involves the specifications of the inter-connection between two equipments having different functions. 2. To connects a PLC with the application device, communications channel, and peripherals through various modules and cables. 3. The method or equipment used to communicate between devices.

interference zone

An area that falls within the work envelope of a robot, in which there is the potential for the robot motion to coincide with the motion of another robot or machine, and for a collision to occur.

interlock

An arrangement whereby the operation of one control or mechanism brings about, or prevents, the operations of another.

interrupt

A break in the normal flow of a system or program that occurs in a way that the flow can be resumed from that point at a later time. Interrupts are initiated by two types of signals: 1. signals originating within the computer system to synchronize the operation of the computer system with the outside

world; 2. signals originating exterior to the computer system to synchronize the operation of the computer system with the outside world.

I/O

Abbreviation for input/output or input/output control.

I/O block

A microprocessor-based, configurable, rugged solid state device to which field I/O devices are attached.

I/O electrical isolation

A method of separating field wiring from logic level circuitry. This is typically done through optical isolation devices.

I/O module

A printed circuit assembly that is the interface between user devices and the Series Six PLC.

I/O scan

A method by which the CPU monitors all inputs and controls all outputs within a prescribed time. A period during which each device on the bus is given a turn to send information and listen to all of the broadcast data on the bus.

ISO

The International Standards Organization that establishes the ISO interface standards.

isolation

1. The ability of a logic circuit having more than one inputs to ensure that each input signal is not affected by any of the others. 2. A method of separating field wiring circuitry from logic level circuitry, typically done optically.

item

1. A category displayed on the teach pendant on a menu. 2. A set of adjacent digits, bits, or characters that is treated as a unit and conveys a single unit of information. 3. Any unique manufactured or purchased part or assembly: end product, assembly, subassembly, component, or raw material.

J**jog coordinate systems**

Coordinate systems that help you to move the robot more effectively for a specific application. These systems include JOINT, WORLD, TOOL, and USER.

JOG FRAME

A jog coordinate system you define to make the robot jog the best way possible for a specific application. This can be different from world coordinate frame.

jogging

Pressing special keys on the teach pendant to move the robot.

jog speed

Is a percentage of the maximum speed at which you can jog the robot.

joint

1. A single axis of rotation. There are up to six joints in a robot arm (P-155 swing arm has 8). 2. A jog coordinate system in which one axis is moved at a time.

JOINT

A motion type in which the robot moves the appropriate combination of axes independently to reach a point most efficiently. (Point to point, non-linear motion).

joint interpolated motion

A method of coordinating the movement of the joints so all joints arrive at the desired location at the same time. This method of servo control produces a predictable path regardless of speed and results in the fastest cycle time for a particular move. Also called joint motion.

K**K**

Abbreviation for kilo, or exactly 1024 in computer jargon. Related to 1024 words of memory.

KAREL

The programming language developed for robots by the FANUC America Corporation.

L**label**

An ordered set of characters used to symbolically identify an instruction, a program, a quantity, or a data area.

LCD

See liquid crystal display.

lead time

The span of time needed to perform an activity. In the production and inventory control context, this activity is normally the procurement of materials and/or products either from an outside supplier or from one's own manufacturing facility. Components of lead time can include order preparation time, queue time, move or transportation time, receiving and inspection time.

LED

See Light Emitting Diode.

LED display

An alphanumeric display that consists of an array of LEDs.

Light Emitting Diode

A solid-state device that lights to indicate a signal on electronic equipment.

limiting device

A device that restricts the work envelope by stopping or causing to stop all robot motion and that is independent of the control program and the application programs.

limit switch

A switch that is actuated by some part or motion of a machine or equipment to alter the electrical circuit associated with it. It can be used for position detection.

linear

A motion type in which the appropriate combination of axes move in order to move the robot TCP in a straight line while maintaining tool center point orientation.

liquid crystal display

A digital display on the teach pendant that consists of two sheets of glass separated by a sealed-in, normally transparent, liquid crystal material. Abbreviated LCD.

little-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

load

1. The weight (force) applied to the end of the robot arm. 2. A device intentionally placed in a circuit or connected to a machine or apparatus to absorb power and convert it into the desired useful form. 3. To copy programs or data into memory storage.

location

1. A storage position in memory uniquely specified by an address. 2. The coordinates of an object used in describing its x, y, and z position in a Cartesian coordinate system.

lockout/tagout

The placement of a lock and/or tag on the energy isolating device (power disconnecting device) in the off or open position. This indicates that the energy isolating device or the equipment being controlled will not be operated until the lock/tag is removed.

log

A record of values and/or action for a given function.

logic

A fixed set of responses (outputs) to various external conditions (inputs). Also referred to as the program.

loop

The repeated execution of a series of instructions for a fixed number of times, or until interrupted by the operator.

M

mA

See milliamperere.

machine language

A language written in a series of bits that are understandable by, and therefore instruct, a computer. This is a "first level" computer language, as compared to a "second level" assembly language, or a "third level" compiler language.

machine lock

A test run option that allows the operator to run a program without having the robot move.

macro

A source language instruction from which many machine-language instructions can be generated.

magnetic disk

A metal or plastic floppy disk (not supported on version 7.10 and later) that looks like a phonograph record whose surface can store data in the form of magnetized spots.

magnetic disk storage

A storage device or system consisting of magnetically coated metal disks.

magnetic tape

Plastic tape, like that used in tape recorder, on which data is stored in the form of magnetized spots.

maintenance

Keeping the robots and system in their proper operating condition.

MC

See memory card.

mechanical unit

The robot arm, including auxiliary axis, and hood/deck and door openers.

medium

plural **media** . The physical substance upon which data is recorded, such as a memory card (or floppy disk which is not supported on version 7.10 and later).

memory

A device or media used to store information in a form that can be retrieved and is understood by the computer or controller hardware. Memory on the controller includes C-MOS RAM, Flash ROM and D-RAM.

memory card

A C-MOS RAM memory card or a flash disk-based PC card.

menu

A list of options displayed on the teach pendant screen.

message

A group of words, variable in length, transporting an item of information.

microprocessor

A single integrated circuit that contains the arithmetic, logic, register, control and memory elements of a computer.

microsecond

One millionth (0.000001) of a second

milliampere

One one-thousandth of an ampere. Abbreviated mA.

millisecond

One thousandth of a second. Abbreviated msec.

module

A distinct and identifiable unit of computer program for such purposes as compiling, loading, and linkage editing. It is eventually combined with other units to form a complete program.

motion type

A feature that allows you to select how you want the robot to move from one point to the next. MOTYPES include joint, linear, and circular.

mode

1. One of several alternative conditions or methods of operation of a device. 2. The most common or frequent value in a group of values.

N**network**

1. The interconnection of a number of devices by data communication facilities. "Local networking" is the communications network internal to a robot. "Global networking" is the ability to provide communications connections outside of the robot's internal system. 2. Connection of geographically separated computers and/or terminals over communications lines. The control of transmission is managed by a standard protocol.

non-volatile memory

Memory capable of retaining its stored information when power is turned off.

O

Obstacle Model.

Used in Interference Checking, the Obstacle Model is the set of virtual model elements (spheres, cylinders, and planes) that are used to represent the shape and the location of a given obstacle in space.

off-line

Equipment or devices that are not directly connected to a communications line.

off-line operations

Data processing operations that are handled outside of the regular computer program. For example, the computer might generate a report off-line while the computer was doing another job.

off-line programming

The development of programs on a computer system that is independent of the "on-board" control of the robot. The resulting programs can be copied into the robot controller memory.

offset

The count value output from a A/D converter resulting from a zero input analog voltage. Used to correct subsequent non-zero measurements also incremental position or frame adjustment value.

on-line

A term to describe equipment or devices that are connected to the communications line.

on-line processing

A data processing approach where transactions are entered into the computer directly, as they occur.

operating system

Lowest level system monitor program.

operating work envelope

The portion of the restricted work envelope that is actually used by the robot while it is performing its programmed motion. This includes the maximum the end-effector, the workpiece, and the robot itself.

operator

A person designated to start, monitor, and stop the intended productive operation of a robot or robot system.

operator box

A control panel that is separate from the robot and is designed as part of the robot system. It consists of the buttons, switches, and indicator lights needed to operate the system.

operator panel

A control panel designed as part of the robot system and consisting of the buttons, switches, and indicator lights needed to operate the system.

optional features

Additional capabilities available at a cost above the base price.

OR

An operation that places two contacts or groups of contacts in parallel. Any of the contacts can control the resultant status, also a mathematical operation.

orientation

The attitude of an object in space. Commonly described by three angles: rotation about x (w), rotation about y (p), and rotation about z (r).

origin

The point in a Cartesian coordinate system where axes intersect; the reference point that defines the location of a frame.

OT

See overtravel.

output

Information that is transferred from the CPU for control of external devices or processes.

output device

A device, such as starter motors, solenoids, that receive data from the programmable controller.

output module

An I/O module that converts logic levels within the CPU to a usable output signal for controlling a machine or process .

outputs

Signals, typically on or off, that controls external devices based upon commands from the CPU.

override

See general override.

overtravel

A condition that occurs when the motion of a robot axis exceeds its prescribed limits.

overwrite

To replace the contents of one file with the contents of another file when copying.

P**parity**

The anticipated state, odd or even, of a set of binary digits.

parity bit

A binary digit added to an array of bits to make the sum of all bits always odd or always even.

parity check

A check that tests whether the number of ones (or zeros) in an array of binary digits is odd or even.

parity error

A condition that occurs when a computed parity check does not agree with the parity bit.

part

A material item that is used as a component and is not an assembly or subassembly.

pascal

A unit of pressure in the meter-kilogram-second system equivalent to one newton per square meter.

path

1. A variable type available in the KAREL system that consists of a list of positions. Each node includes positional information and associated data. 2. The trajectory followed by the TCP in a move.

PCB

See printed circuit board.

PC Interface

The PC Interface option provides the RPC functions and PC send macros required by applications created using PC Developer's Kit.

pendant

See teach pendant.

PLC

See programmable logic controller or cell controller.

PMC

The programmable machine controller (PMC) functions provide a ladder logic programming environment to create PMC functions. This provides the capability to use the robot I/O system to run PLC programs in the background of normal robot operations. This function can be used to control bulk supply systems, fixed automation that is part of the robot workcell, or other devices that would normally require basic PLC controls.

printed circuit board

A flat board whose front contains slots for integrated circuit chips and connections for a variety of electronic components, and whose back is printed with electrically conductive pathways between the components.

production mode

See automatic mode.

program

1. A plan for the solution of a problem. A complete program includes plans for the transcription of data, coding for the computer, and plans for the absorption of the results into the system. 2. A sequence of instructions to be executed by the computer or controller to control a robot/robot system. 3. To furnish a computer with a code of instructions. 4. To teach a robot system a specific set of movements and instructions to do a task.

programmable controller

See programmable logic controller or cell controller.

programmable logic controller

A solid-state industrial control device that receives inputs from user-supplied control devices, such as switches and sensors, implements them in a precise pattern determined by ladder diagram-based programs stored in the user memory, and provides outputs for control of processes or user-supplied devices such as relays and motor starters.

Program ToolBox

The Program ToolBox software provides programming utilities such as mirror image and flip wrist editing capabilities.

protocol

A set of hardware and software interfaces in a terminal or computer that allows it to transmit over a communications network, and that collectively forms a communications language.

psi

Pounds per square inch.

Q**queue.**

1. Waiting lines resulting from temporary delays in providing service. 2. The amount of time a job waits at a work center before set-up or work is performed on the job. See also job queue.

R**RAM**

See Random Access Memory.

random access

A term that describes files that do not have to be searched sequentially to find a particular record but can be addressed directly.

Random Access Memory

1. Volatile, solid-state memory used for storage of programs and locations; battery backup is required. 2. The working memory of the controller. Programs and variable data must be loaded into RAM before the program can execute or the data can be accessed by the program.

range

1. A characterization of a variable or function. All the values that a function can possess. 2. In statistics, the spread in a series of observations. 3. A programmable voltage or current spectrum of values to which input or output analog signals can be limited.

RI

Robot input.

RO

Robot output.

read

To copy, usually from one form of storage to another, particularly from external or secondary storage to internal storage. To sense the meaning of arrangements of hardware. To sense the presence of information on a recording medium.

Read Only Memory

A digital memory containing a fixed pattern of bits that you cannot alter.

record

To store the current set or sets of information on a storage device.

recovery

The restoration of normal processing after a hardware or software malfunction through detailed procedures for file backup, file restoration, and transaction logging.

register

1. A special section of primary storage in a computer where data is held while it is being worked on.
2. A memory device capable of containing one or more computer bits or words.

remote/local

A device connection to a given computer, with remote devices being attached over communications lines and local devices attached directly to a computer channel; in a network, the computer can be a remote device to the CPU controlling the network.

repair

To restore robots and robot systems to operating condition after damage, malfunction, or wear.

repeatability

The closeness of agreement among the number of consecutive movements made by the robot arm to a specific point.

reset

To return a register or storage location to zero or to a specified initial condition.

restricted work envelope

That portion of the work envelope to which a robot is restricted by limiting devices that establish limits that will not be exceeded in the event of any reasonably foreseeable failure of the robot or its controls. The maximum distance the robot can travel after the limited device is actuated defines the restricted work envelope of the robot.

RIA

Robotic Industries Association Subcommittee of the American National Standards Institute, Inc.

robot

A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices, through variable programmed motions for the performance of a variety of tasks.

Robot Model.

Used in Interference Checking, the Robot Model is the set of virtual model elements (sphere and cylinders) that are used to represent the location and shape of the robot arm with respect to the robot's base. Generally, the structure of a six axes robot can be accurately modeled as a series of cylinders and spheres. Each model element represents a link or part of the robot arm.

ROM

See Read Only Memory.

routine

1. A list of coded instructions in a program. 2. A series of computer instructions that performs a specific task and can be executed as often as needed during program execution.

S**saving data.**

Storing program data in Flash ROM, to a floppy disk (not supported on version 7.10 and later), or memory card.

scfm

Standard cubic feet per minute.

scratch start

Allows you to enable and disable the automatic recovery function.

sensor

A device that responds to physical stimuli, such as heat, light, sound pressure, magnetism, or motion, and transmits the resulting signal or data for providing a measurement, operating a control or both. Also a device that is used to measure or adjust differences in voltage in order to control sophisticated machinery dynamically.

serial communication

A method of data transfer within a PLC whereby the bits are handled sequentially rather than simultaneously as in parallel transmission.

serial interface

A method of data transmission that permits transmitting a single bit at a time through a single line. Used where high speed input is not necessary.

Server Side Include (SSI)

A method of calling or "including" code into a web page.

servomotor

An electric motor that is controlled to produce precision motion. Also called a "smart" motor.

SI

System input.

signal

The event, phenomenon, or electrical quantity that conveys information from one point to another.

significant bit

A bit that contributes to the precision of a number. These are counted starting with the bit that contributes the most value, of "most significant bit", and ending with the bit that contributes the least value, or "least significant bit".

singulating

Separating parts into a single layer.

slip sheet

A sheet of material placed between certain layers of a unit load. Also known as tier sheet.

SO

System output.

specific gravity

The ratio of a mass of solid or liquid to the mass of an equal volume of water at 45C. You must know the specific gravity of the dispensing material to perform volume signal calibration. The specific gravity of a dispensing material is listed on the MSDS for that material.

SRAM

A read/write memory in which the basic memory cell is a transistor. SRAM (or S-RAM) tends to have a lower density than DRAM. A battery can be used to retain the content upon loss of power.

slpm

Standard liters per minute.

Standard Operator Panel (SOP).

A panel that is made up of buttons, keyswitches, and connector ports.

state

The on or off condition of current to and from an input or output device.

statement

See instruction.

storage device

Any device that can accept, retain, and read back one or more times. The available storage devices are SRAM, Flash ROM (FROM or F-ROM), floppy disks (not available on version 7.10 and later), memory cards, or a USB memory stick.

system variable

An element that stores data used by the controller to indicate such things as robot specifications, application requirements, and the current status of the system.

T**Tare**

The difference between the gross weight of an object and its contents, and the object itself. The weight of an object without its contents.

TCP

See tool center point.

teaching

Generating and storing a series of positional data points effected by moving the robot arm through a path of intended motions.

teach mode

1. The mode of operation in which a robot is instructed in its motions, usually by guiding it through these motions using a teach pendant. 2. The generation and storage of positional data. Positional data can be taught using the teach pendant to move the robot through a series of positions and recording those positions for use by an application program.

teach pendant

1. A hand-held device used to instruct a robot, specifying the character and types of motions it is to undertake. Also known as teach box, teach gun. 2. A portable device, consisting of an LCD display and a keypad, that serves as a user interface to the KAREL system and attaches to the operator box or operator panel via a cable. The teach pendant is used for robot operations such as jogging the robot, teaching and recording positions, and testing and debugging programs.

telemetry

The method of transmission of measurements made by an instrument or a sensor to a remote location.

termination type

Feature that controls the blending of robot motion between segments.

tool

A term used loosely to define something mounted on the end of the robot arm, for example, a hand, gripper, or an arc welding torch.

tool center point

1. The location on the end-effector or tool of a robot hand whose position and orientation define the coordinates of the controlled object. 2. Reference point for position control, that is, the point on the tool that is used to teach positions. Abbreviated TCP.

TOOL Frame

The Cartesian coordinate system that has the position of the TCP as its origin to set. The z-axis of the tool frame indicates the approach vector for the tool.

TP.

See teach pendant.

transducer

A device for converting energy from one form to another.

U**UOP**

See user operator panel.

URL

Universal Resource Locator. A standard addressing scheme used to locate or reference files on web servers.

USB memory stick

The controller USB memory stick interface supports a USB 1.1 interface. The USB Organization specifies standards for USB 1.1 and 2.0. Most memory stick devices conform to the USB 2.0 specification for operation and electrical standards. USB 2.0 devices as defined by the USB Specification must be backward compatible with USB 1.1 devices. However, FANUC America Corporation does not support any security or encryption features on USB memory sticks. The controller supports most widely-available USB Flash memory sticks from 32MB up to 1GB in size.

USER Frame

The Cartesian coordinate system that you can define for a specific application. The default value of the User Frame is the World Frame. All positional data is recorded relative to User Frame.

User Operator Panel

User-supplied control device used in place of or in parallel with the operator panel or operator box supplied with the controller. Abbreviated UOP .

V**variable**

A quantity that can assume any of a given set of values.

variance

The difference between the expected (or planned) and the actual, also statistics definitions.

vision system

A device that collects data and forms an image that can be interpreted by a robot computer to determine the position or to “see” an object.

volatile memory

Memory that will lose the information stored in it if power is removed from the memory circuit device.

W**web server**

An application that allows you to access files on the robot using a standard web browser.

warning device

An audible or visible device used to alert personnel to potential safety hazards.

work envelope

The volume of space that encloses the maximum designed reach of the robot manipulator including the end effector, the workpiece, and the robot itself. The work envelope can be reduced or restricted by limiting devices. The maximum distance the robot can travel after the limit device is actuated is considered the basis for defining the restricted work envelope.

write

To deliver data to a medium such as storage.

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